

Lecture 11: Data Visualization Continued

Heather Mattie, PhD

October 6, 2026



Recipe of the Day!

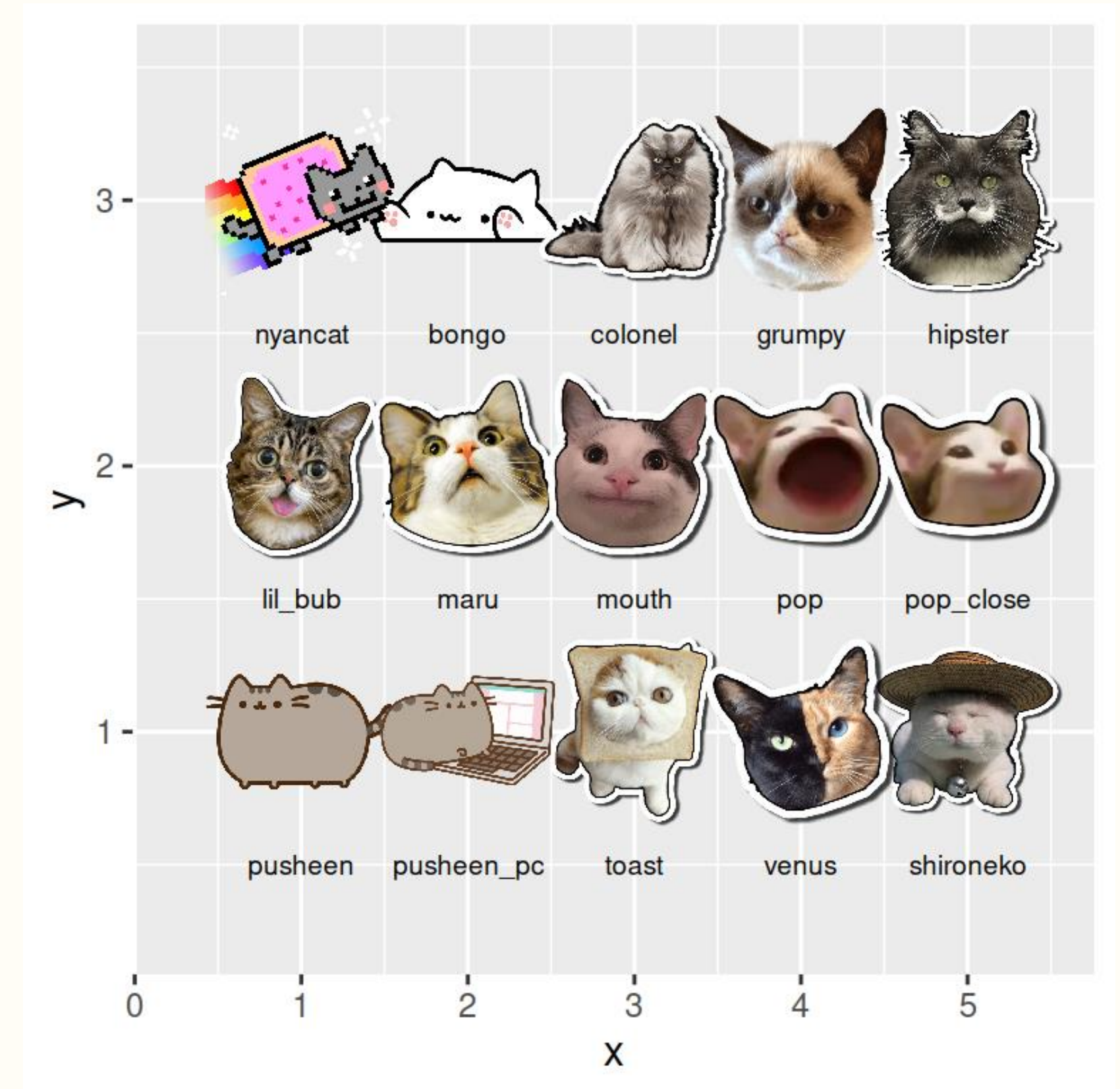
Spaghetti Carbonara



Emre says to start Halloween
costume shopping now!

Announcements

- Heather's office hour: Thursday 12-1pm
- Homework #2 due Friday, October 9 by 11:59pm
- Lab will be held this week



ggcats and geom_cat

Coding question of the day

Using the `gapminder` dataset and `ggplot2`, for the year 1996, create a scatter plot with `fertility` on the x-axis and `life_expectancy` on the y-axis. Color the points according to `continent`, and make the size of the points correspond to the population of the country. Update the `x` and `y` axis labels to make them more informative.

Hint: the arguments `color` and `size` will be useful inside of `ggplot`. Note: you will most likely get ugly legends - don't worry about that right now.

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data(gapminder)
```

Data visualization case study

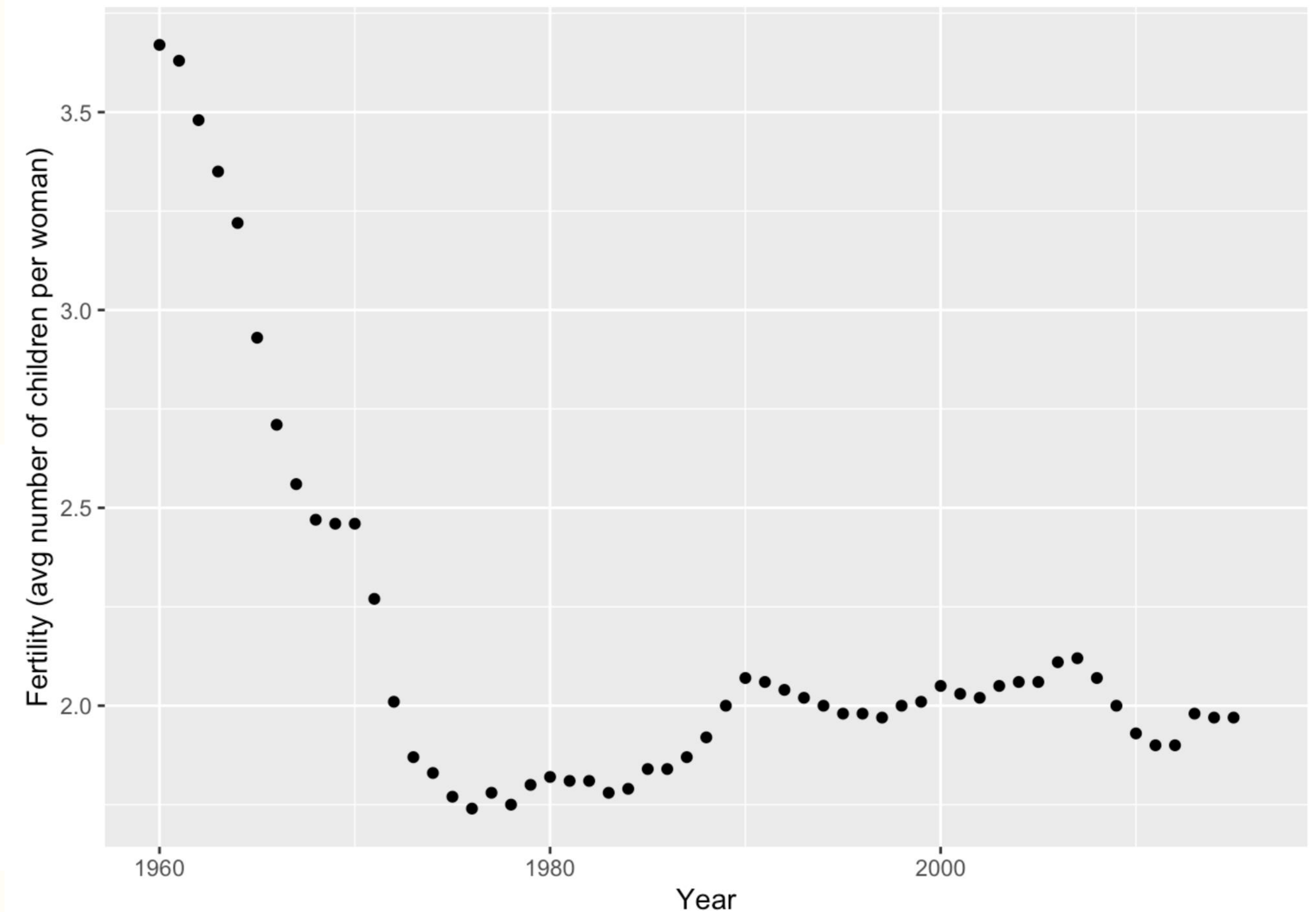
Time series plots

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Time series plots

Time series plots have time on the x-axis and an outcome or measurement of interest on the y-axis

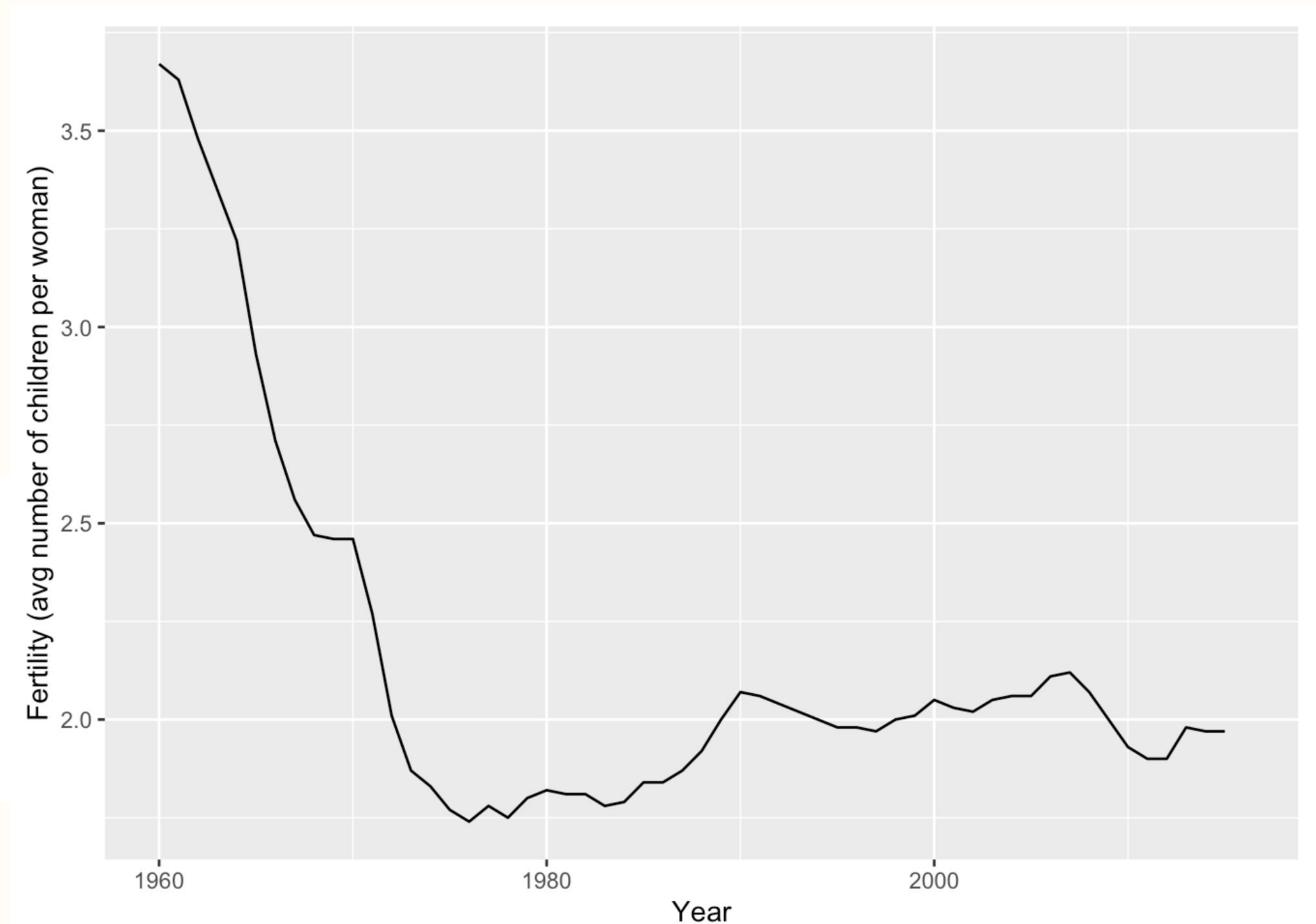
```
library(dslabs)
library(tidyverse)
gapminder %>% filter(country == "United States") %>%
  ggplot(aes(year, fertility)) +
  geom_point() +
  xlab("Year") +
  ylab("Fertility (avg number of children per woman)")
```



Time series plots

Time series plots have time on the x-axis and an outcome or measurement of interest on the y-axis

```
gapminder %>% filter(country == "United States") %>%  
  ggplot(aes(year, fertility)) +  
  geom_line() +  
  xlab("Year") +  
  ylab("Fertility (avg number of children per woman)")
```

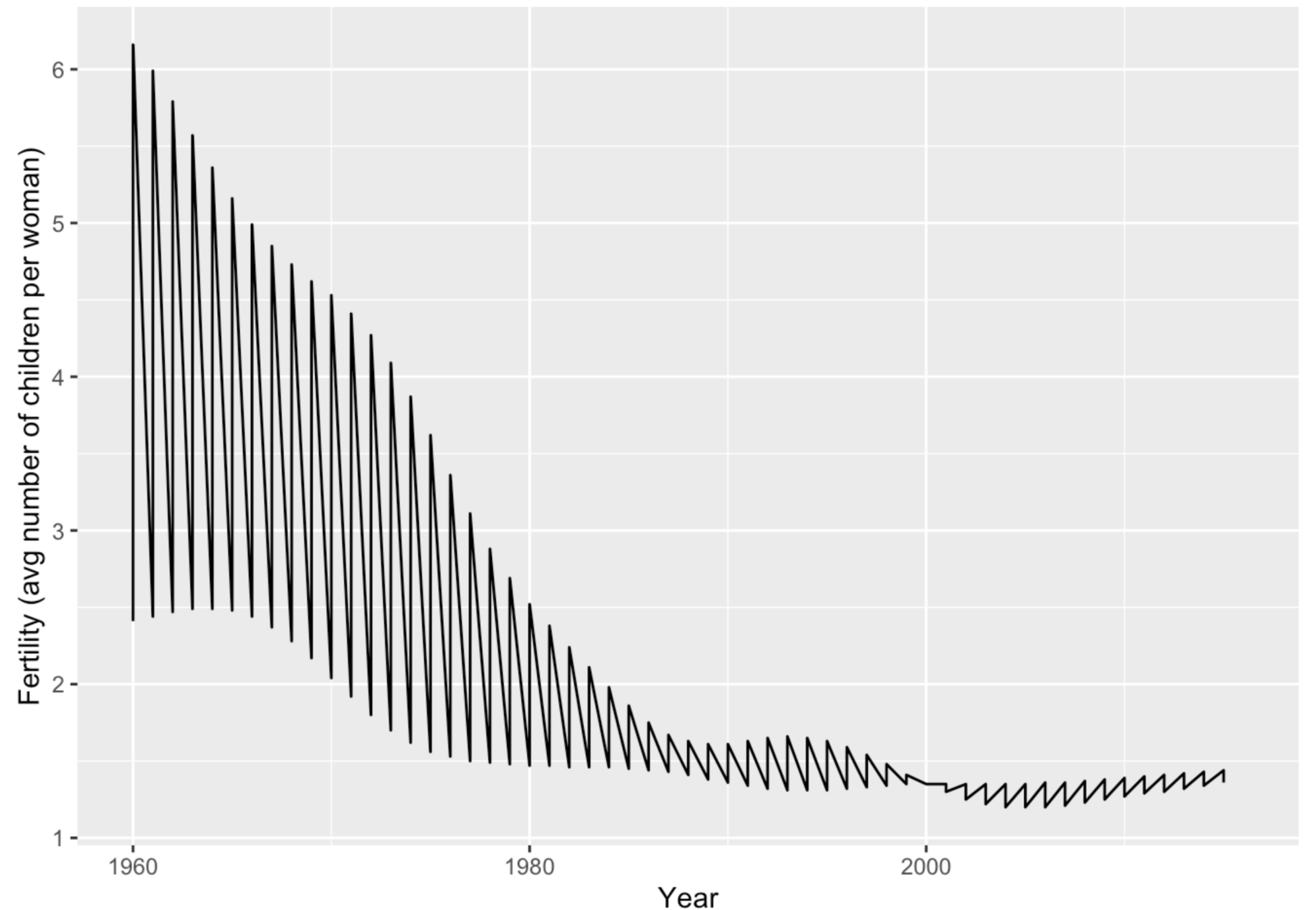


Time series plots

```
countries <- c("South Korea", "Germany")
gapminder %>% filter(country %in% countries) %>%
  ggplot(aes(year, fertility)) +
  geom_line() +
  xlab("Year") +
  ylab("Fertility (avg number of children per woman)")
```

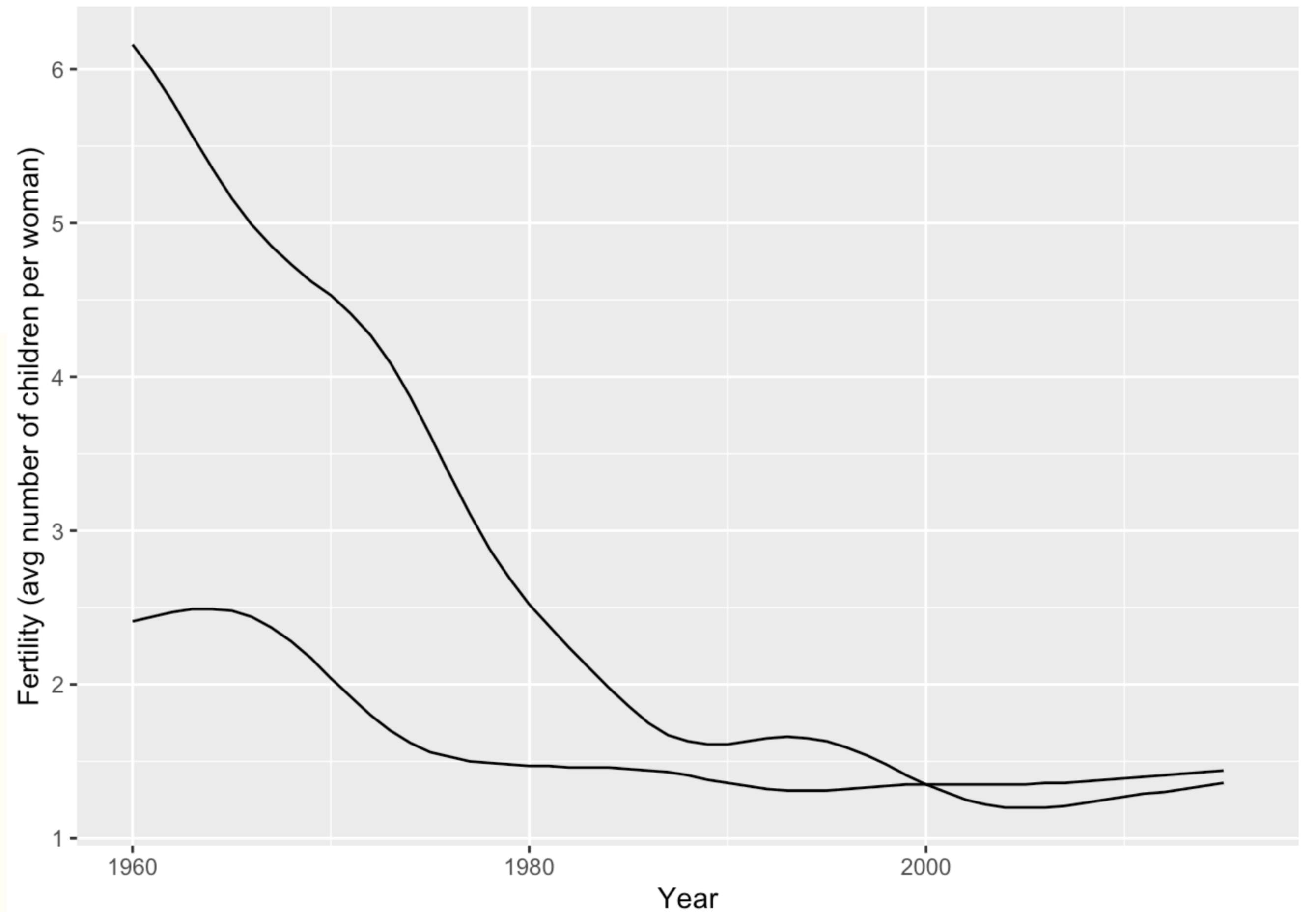
Time series plots

```
countries <- c("South Korea", "Germany")
gapminder %>% filter(country %in% countries) %>%
  ggplot(aes(year, fertility)) +
  geom_line() +
  xlab("Year") +
  ylab("Fertility (avg number of children per woman)")
```



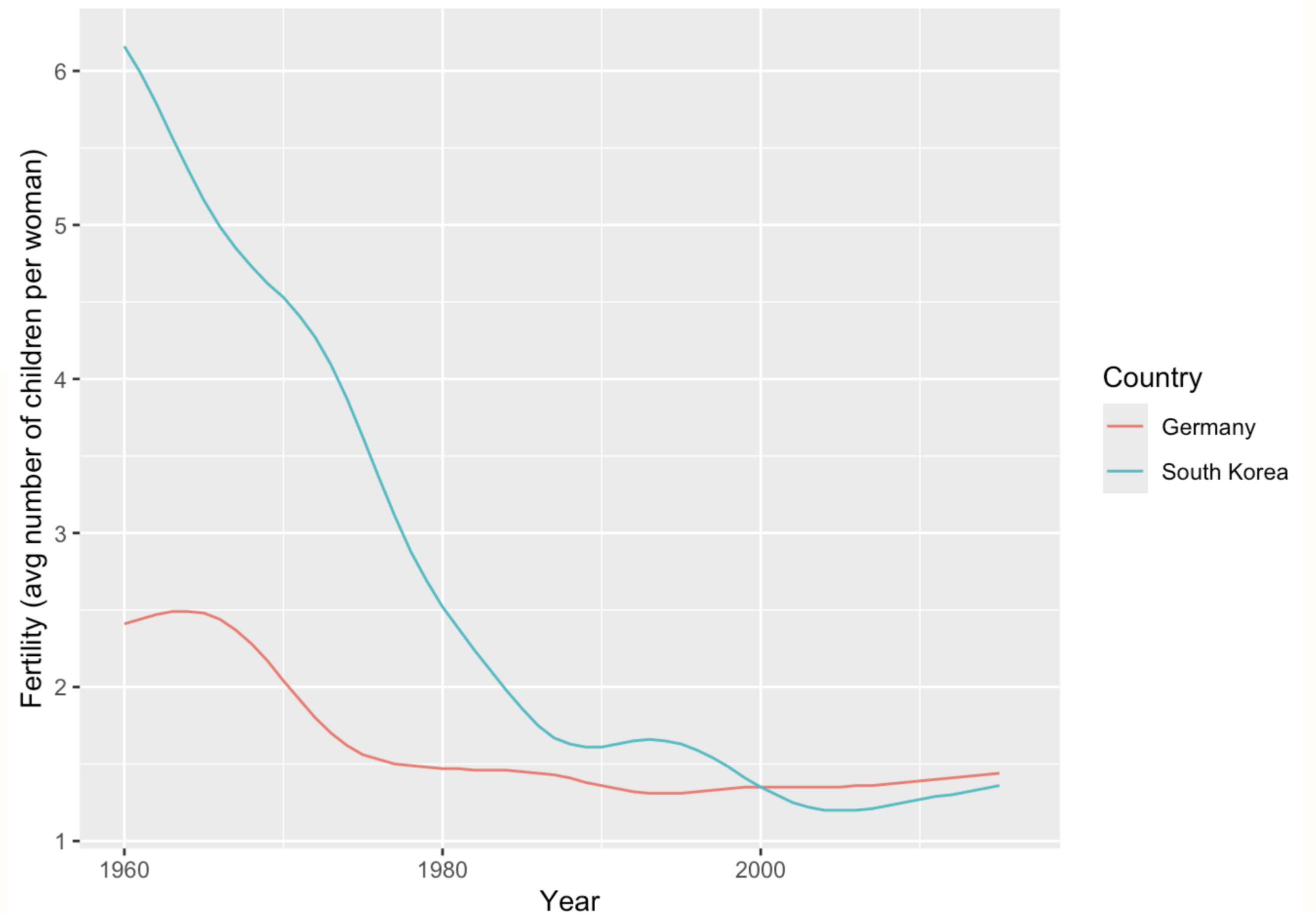
Time series plots

```
countries <- c("South Korea", "Germany")
gapminder %>% filter(country %in% countries) %>%
  ggplot(aes(year, fertility, group = country)) +
  geom_line() +
  xlab("Year") +
  ylab("Fertility (avg number of children per woman)")
```



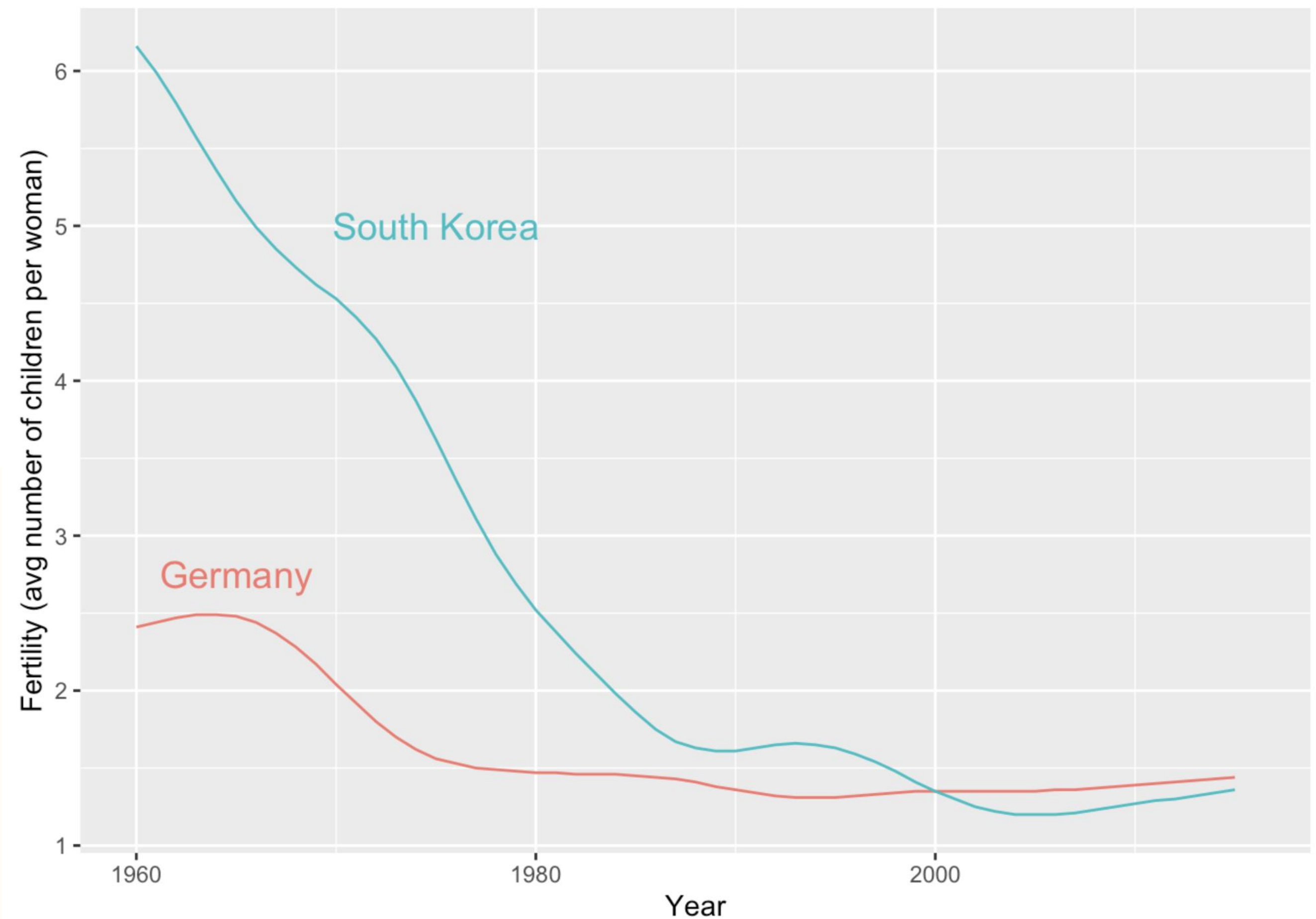
Time series plots

```
countries <- c("South Korea", "Germany")
gapminder %>% filter(country %in% countries) %>%
  ggplot(aes(year, fertility, color = country)) +
  geom_line() +
  xlab("Year") +
  ylab("Fertility (avg number of children per woman)") +
  scale_color_discrete(name = "Country")
```



Labels over legends

```
labels <- data.frame(country = countries,  
                     x = c(1975, 1965),  
                     y = c(5, 2.75))  
  
gapminder %>% filter(country %in% countries) %>%  
  ggplot(aes(year, fertility, color = country)) +  
  geom_line() +  
  geom_text(data = labels, aes(x, y, label = country), size = 5) +  
  theme(legend.position = "none") +  
  xlab("Year") +  
  ylab("Fertility (avg number of children per woman)")
```



Summary

- Time series plots are line plots that show trends over time
- The **group** or **color** arguments must be defined to plot one line per group or category
- Using different colors for different groups or categories eases interpretability
- When only a few lines are plotted, we prefer labels over a legend

Data visualization case study

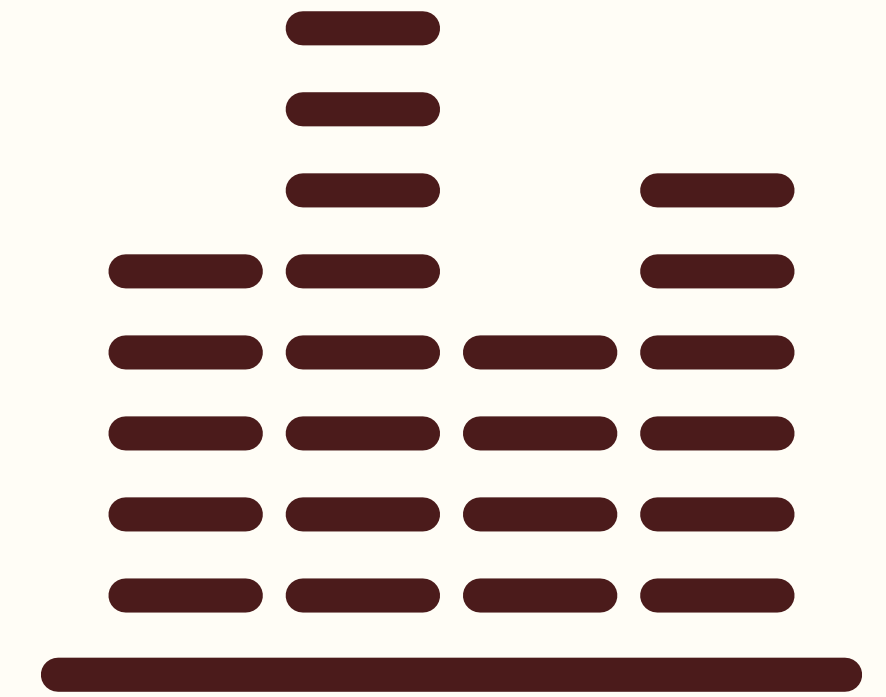
Transformations

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Transformations

Data transformations and axis transformations can make it easier to understand patterns, relationships, and distributions, making a visualization **easier to interpret**

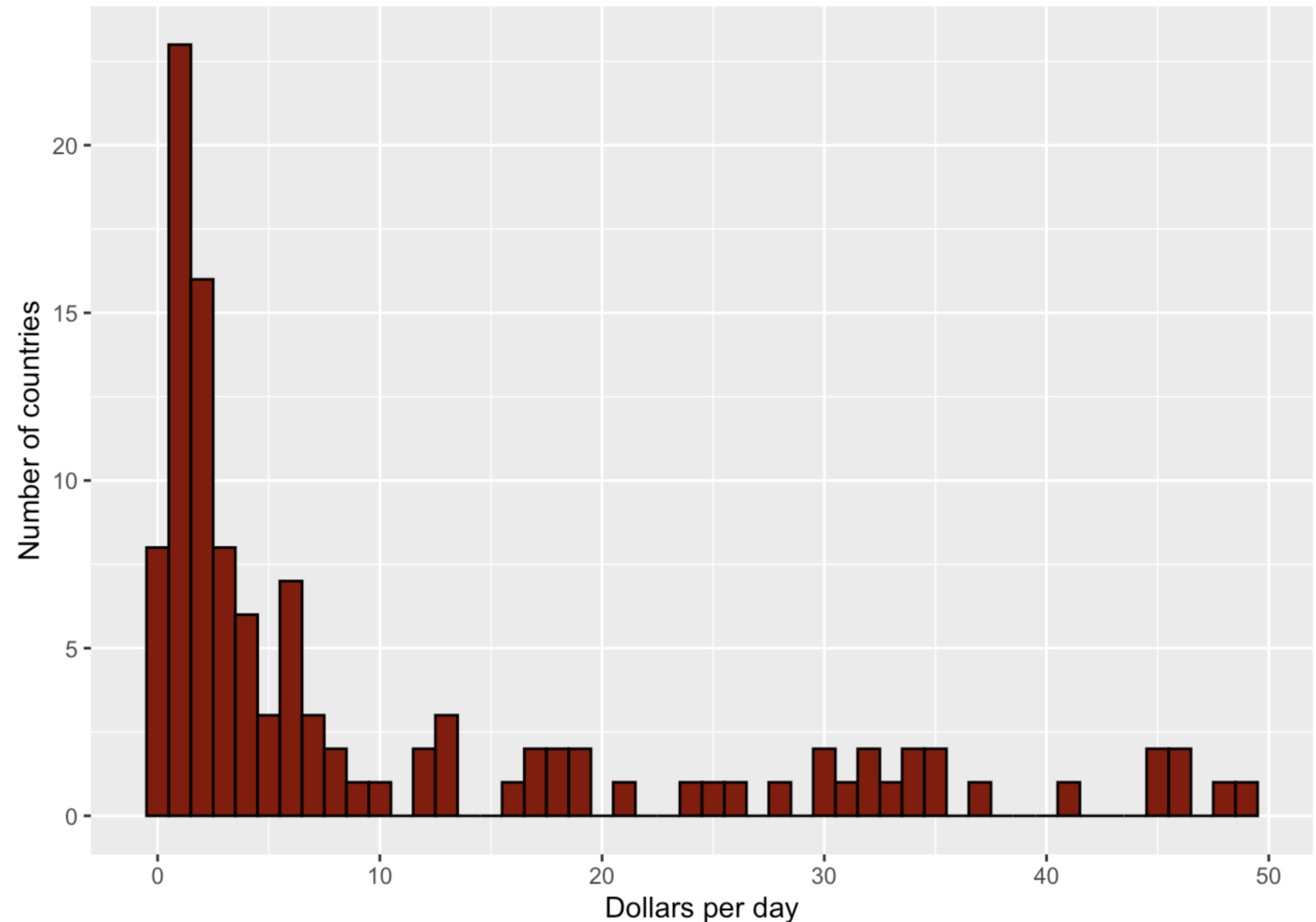
- Data transformations involve mathematical operations on the data
- Axis transformations change the scale and values that appear on a plot axis



Country income distribution

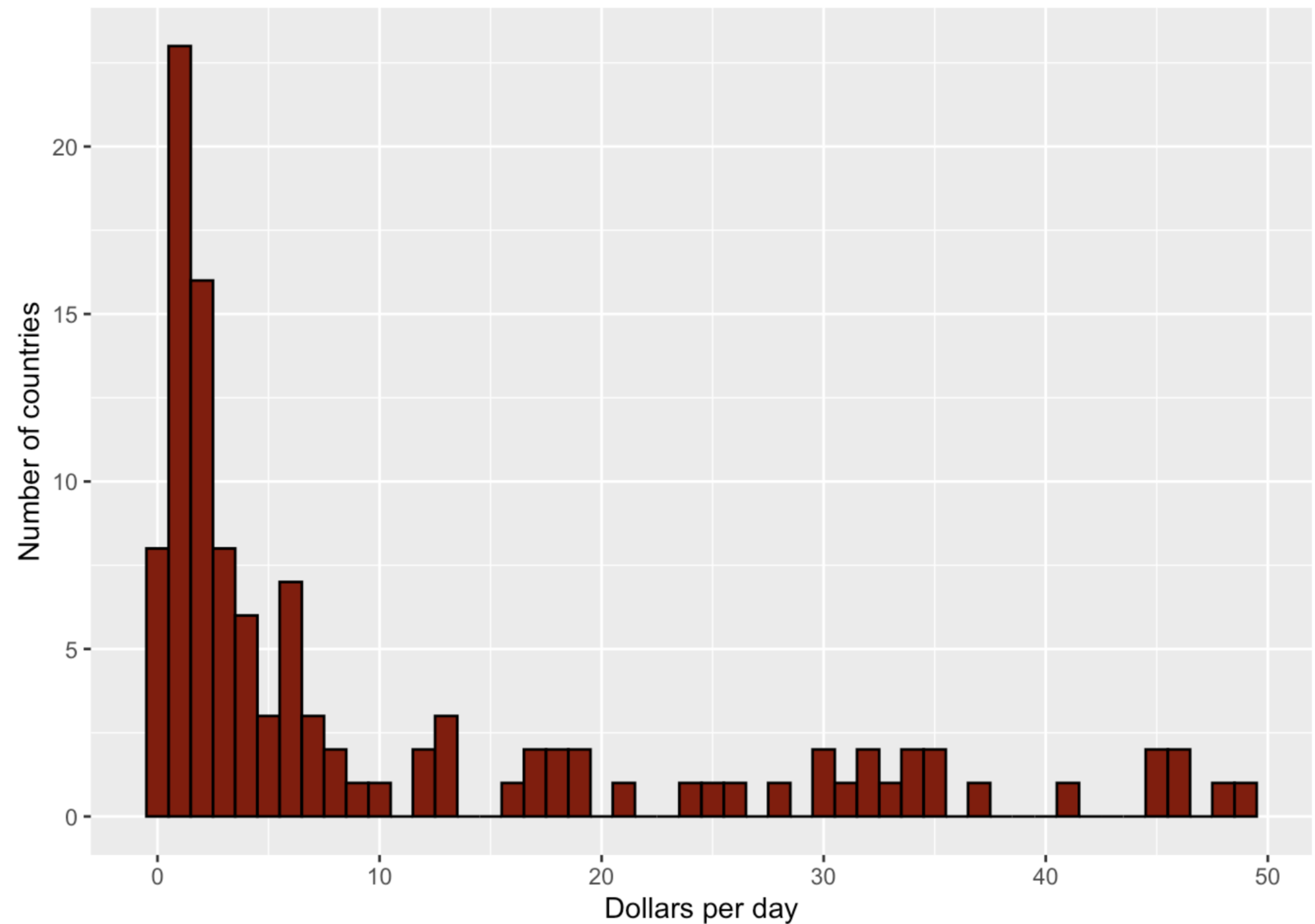
Let's visualize the distribution of country incomes using the **gapminder** data

- The **gdp** column contains the total **gdp** in dollars for a particular country in a particular year
- The values have been adjusted for inflation so comparisons can be made across years



Country income distribution

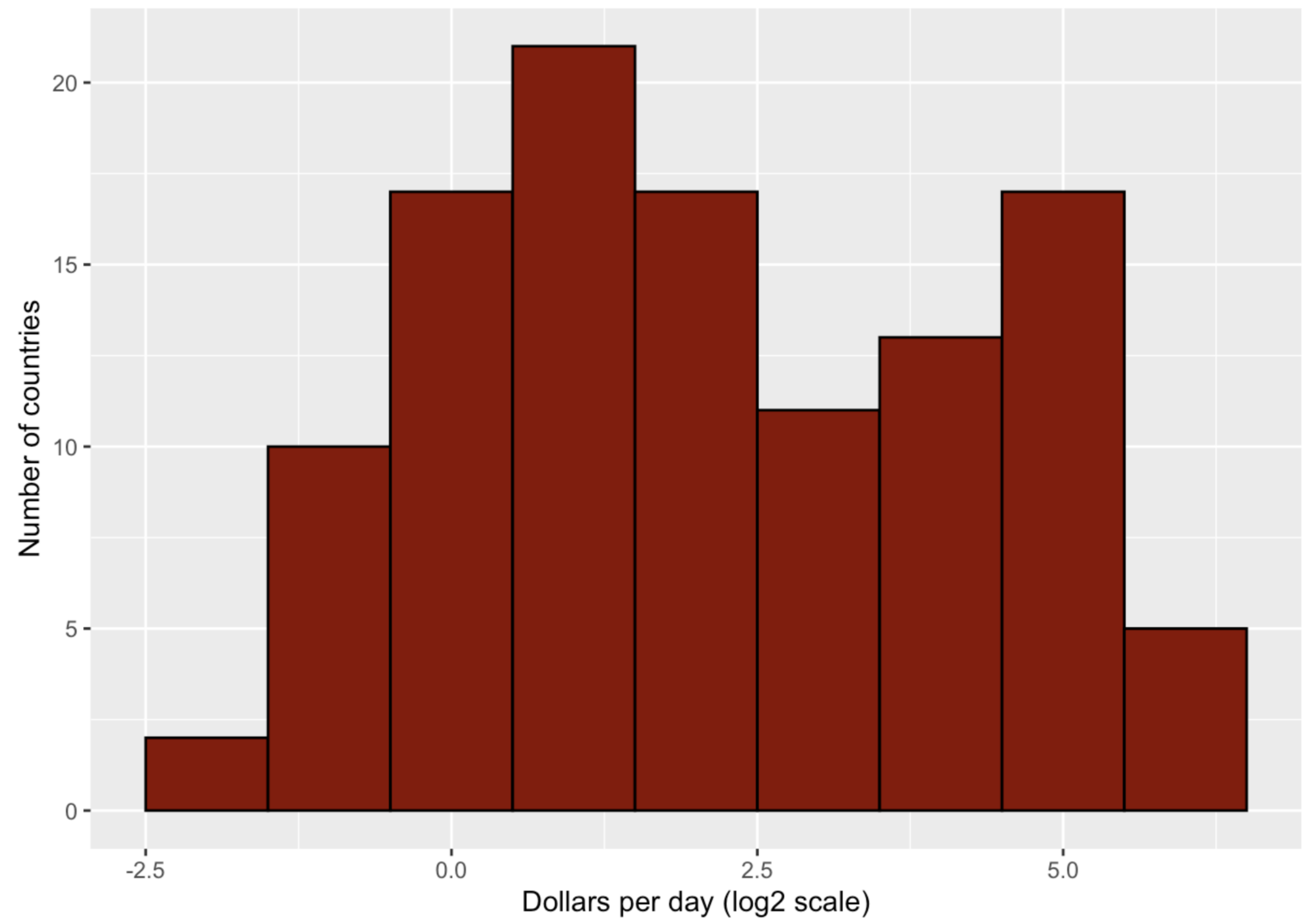
```
gapminder <- gapminder %>%  
  mutate(dollars_per_day = gdp/population/365)  
  
past_year <- 1970  
  
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_histogram(binwidth = 1, color = "black", fill = "darkred") +  
  xlab("Dollars per day") +  
  ylab("Number of countries")
```



Data transformation

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  ggplot(aes(log2(dollars_per_day))) +  
  geom_histogram(binwidth = 1, color = "black", fill = "darkred") +  
  xlab("Dollars per day (log2 scale)") +  
  ylab("Number of countries")
```

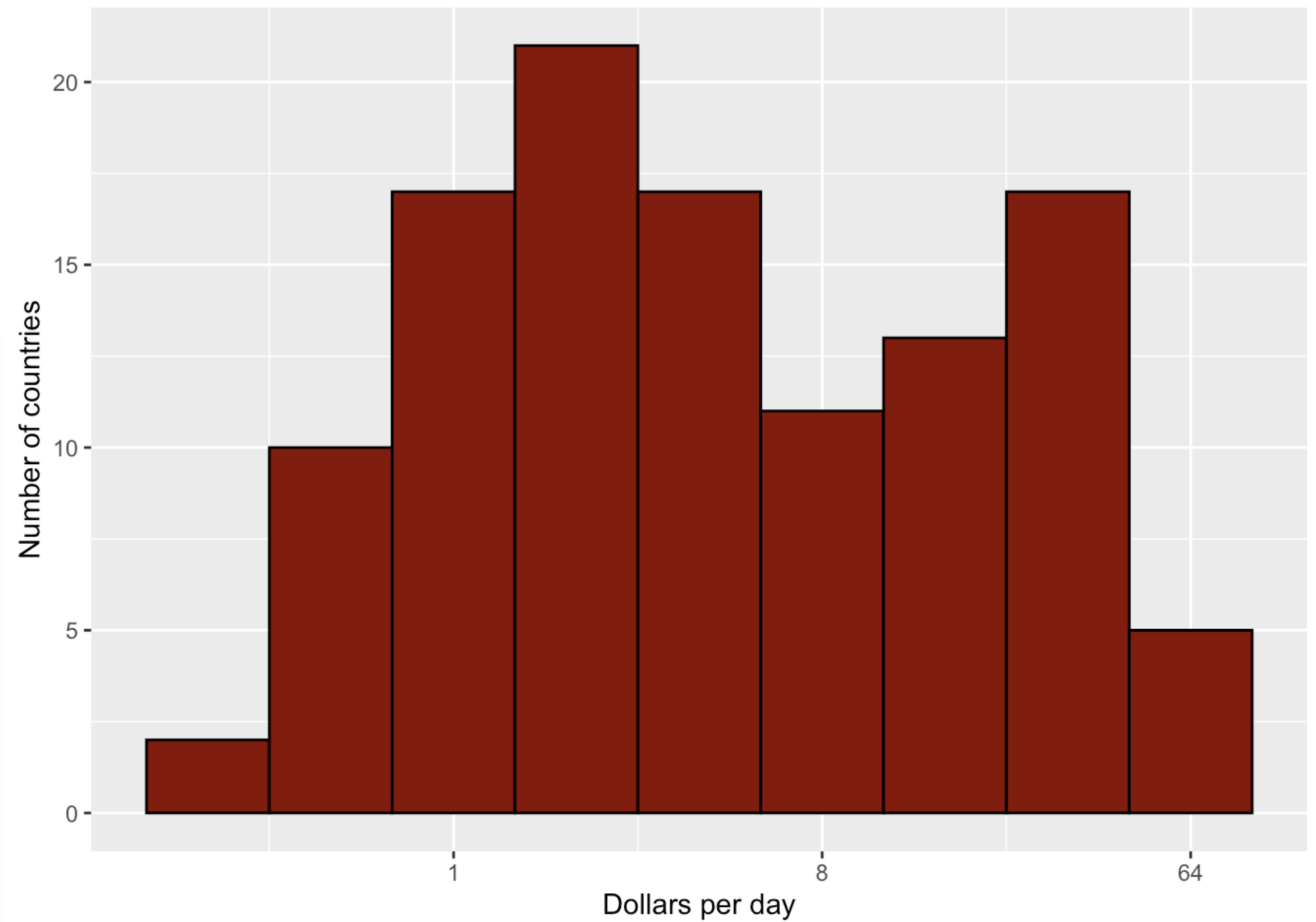
Data transformations change
the values of the data directly



Axis transformation

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_histogram(binwidth = 1, color = "black", fill = "darkred") +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  ylab("Number of countries")
```

Transforming an axis changes the values and scale of an axis but plots the original data values, which is usually easier to interpret



Summary

- Data and axis transformations can improve data visualizations and make them easier to interpret
- Data transformations change the values of the data
- Axis transformations change the values and scale of a plot axis

Comparing Distributions

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Comparing Distributions Roadmap

1. Reordering
2. Smooth density plots

Comparing Distributions

Reordering

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Gapminder regions

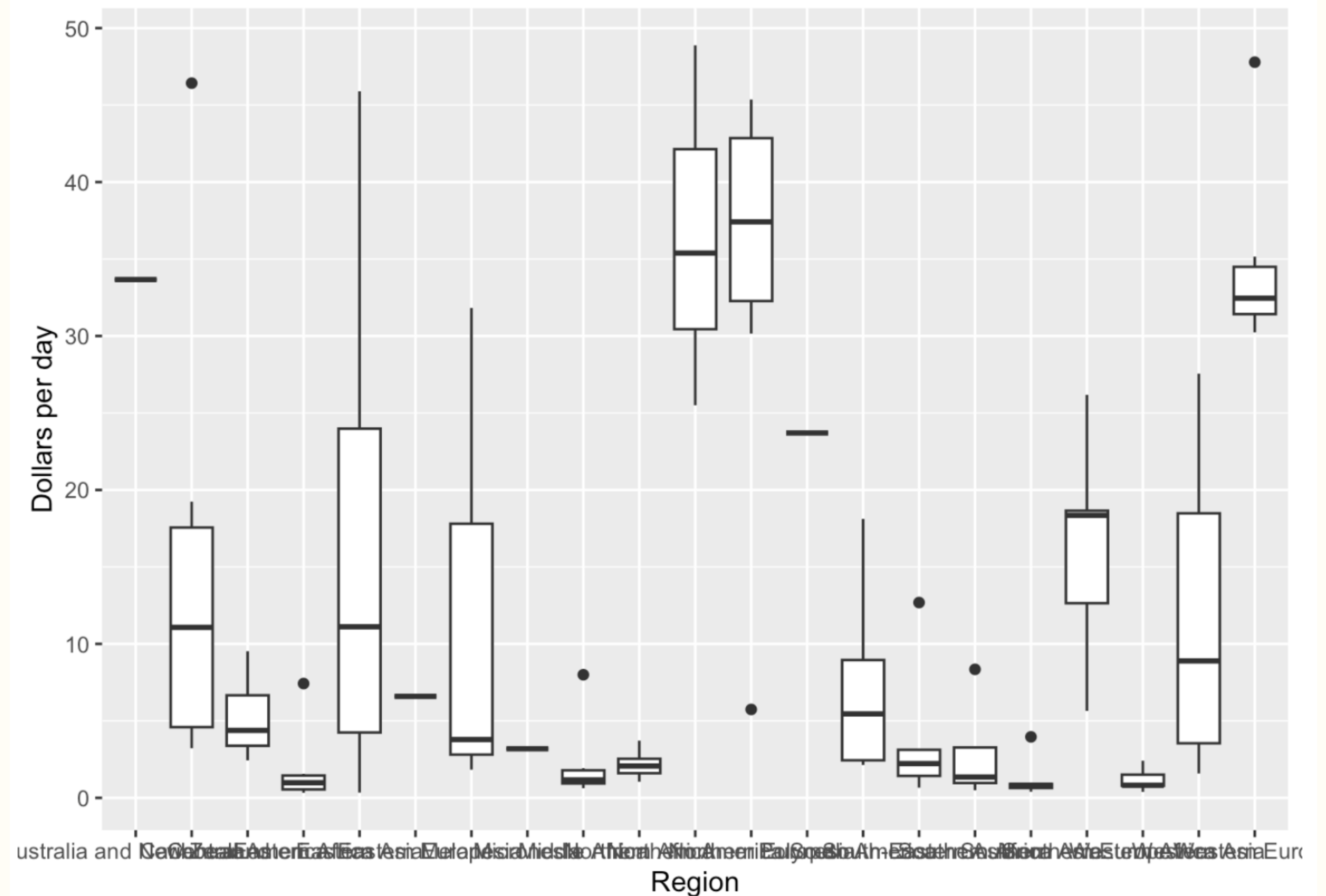
```
levels(gapminder$region)
```

```
## [1] "Australia and New Zealand" "Caribbean"  
## [3] "Central America"          "Central Asia"  
## [5] "Eastern Africa"            "Eastern Asia"  
## [7] "Eastern Europe"            "Melanesia"  
## [9] "Micronesia"                "Middle Africa"  
## [11] "Northern Africa"           "Northern America"  
## [13] "Northern Europe"           "Polynesia"  
## [15] "South America"             "South-Eastern Asia"  
## [17] "Southern Africa"           "Southern Asia"  
## [19] "Southern Europe"           "Western Africa"  
## [21] "Western Asia"              "Western Europe"
```

```
length(levels(gapminder$region))
```

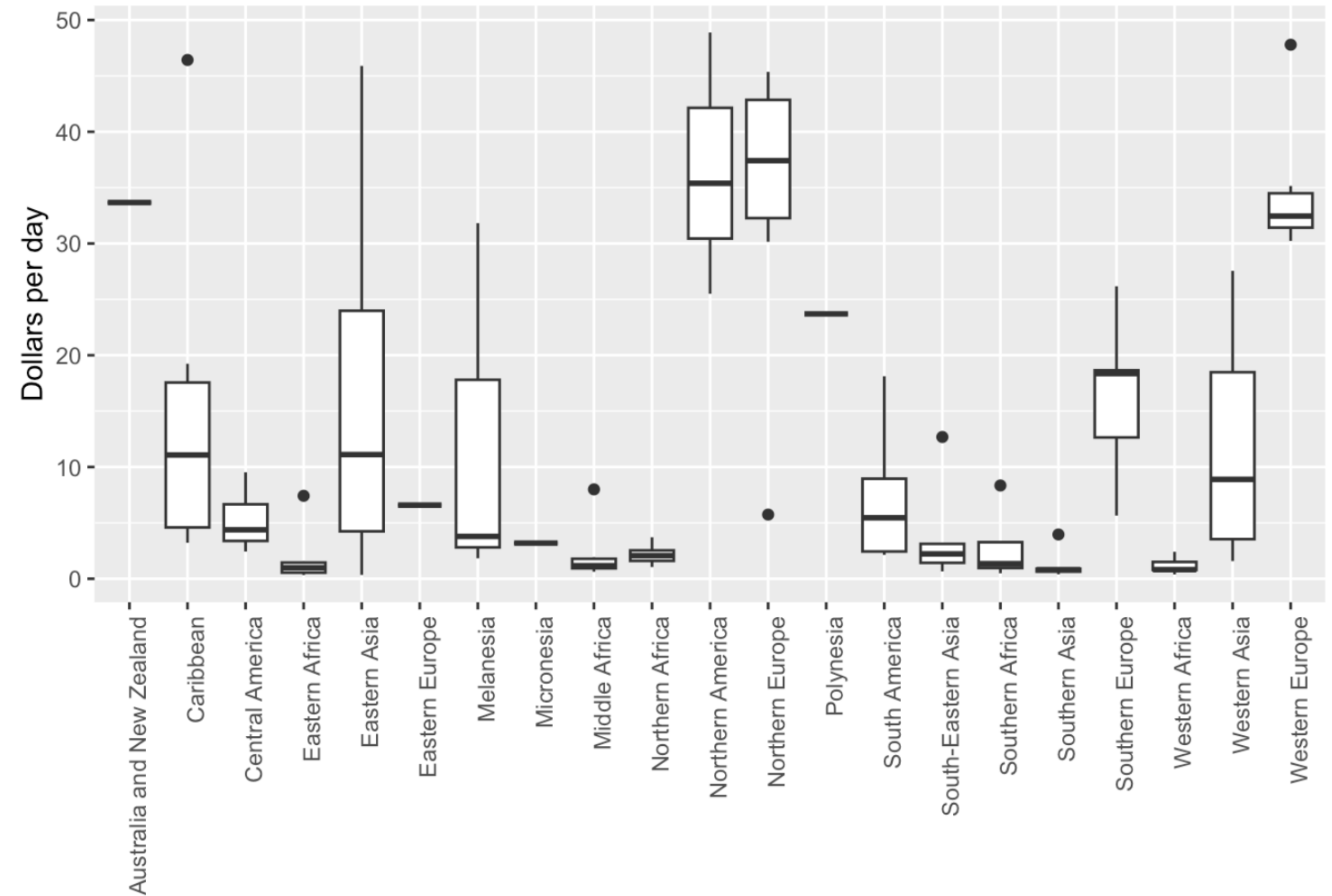
```
## [1] 22
```

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  ggplot(aes(region, dollars_per_day)) +  
  geom_boxplot() +  
  xlab("Region") +  
  ylab("Dollars per day")
```



Vertical axis labels

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  ggplot(aes(region, dollars_per_day)) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  xlab("") +  
  ylab("Dollars per day")
```



reorder

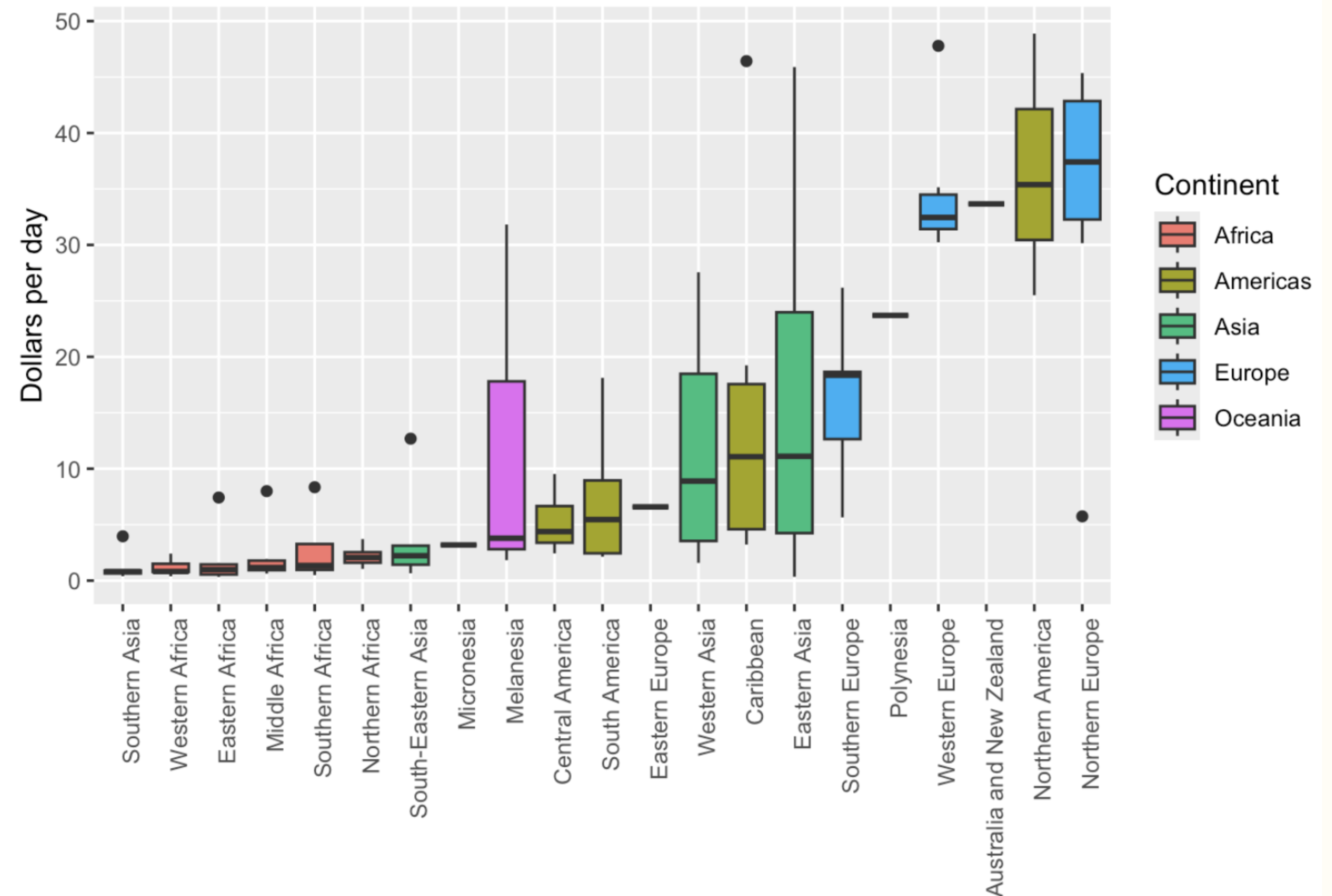
```
fac <- factor(c("Asia", "Asia", "West", "West", "West"))  
levels(fac)
```

```
## [1] "Asia" "West"
```

```
value <- c(10, 11, 12, 6, 4)  
fac <- reorder(fac, value, FUN = mean)  
levels(fac)
```

```
## [1] "West" "Asia"
```

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%  
  ggplot(aes(region, dollars_per_day, fill = continent)) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  xlab("") +  
  ylab("Dollars per day") +  
  labs(fill = "Continent")
```



reorder

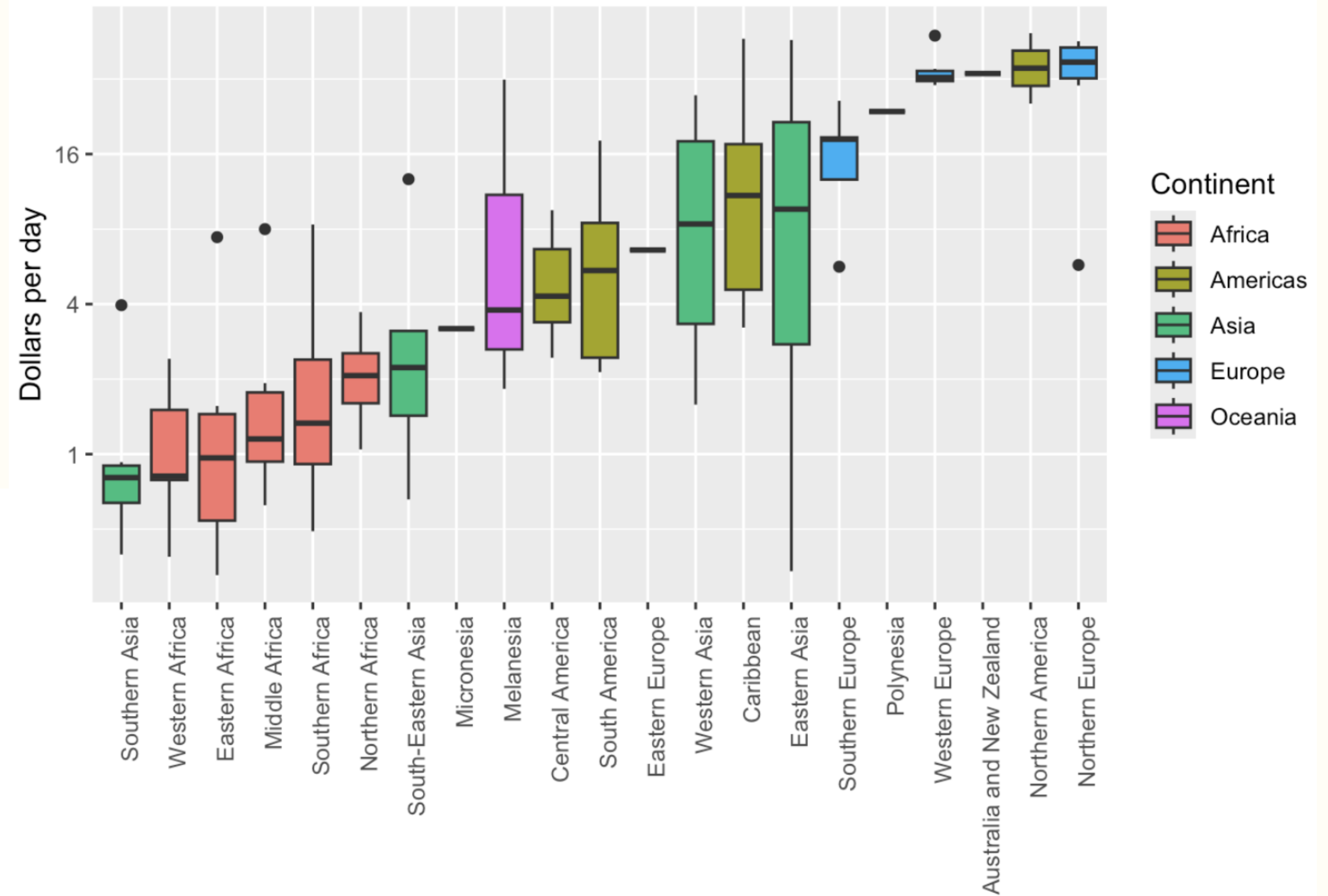
```
fac <- factor(c("Asia", "Asia", "West", "West", "West"))
levels(fac)
```

```
## [1] "Asia" "West"
```

```
value <- c(10, 11, 12, 6, 4)
fac <- reorder(fac, value, FUN = mean)
levels(fac)
```

```
## [1] "West" "Asia"
```

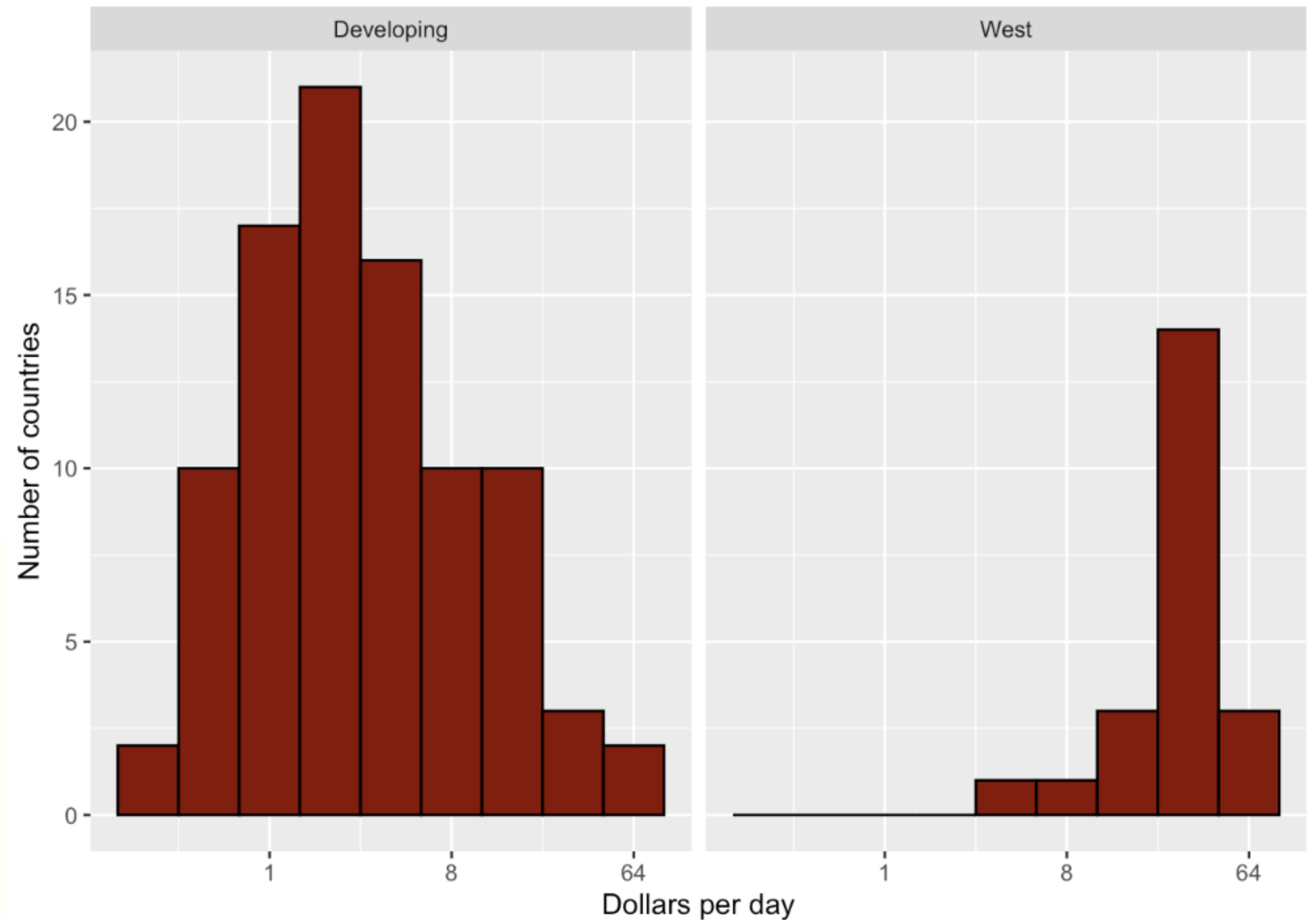
```
gapminder %>%
  filter(year == past_year & !is.na(gdp)) %>%
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%
  ggplot(aes(region, dollars_per_day, fill = continent)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  xlab("") +
  ylab("Dollars per day") +
  labs(fill = "Continent") +
  scale_y_continuous(trans = "log2")
```



Compare across years

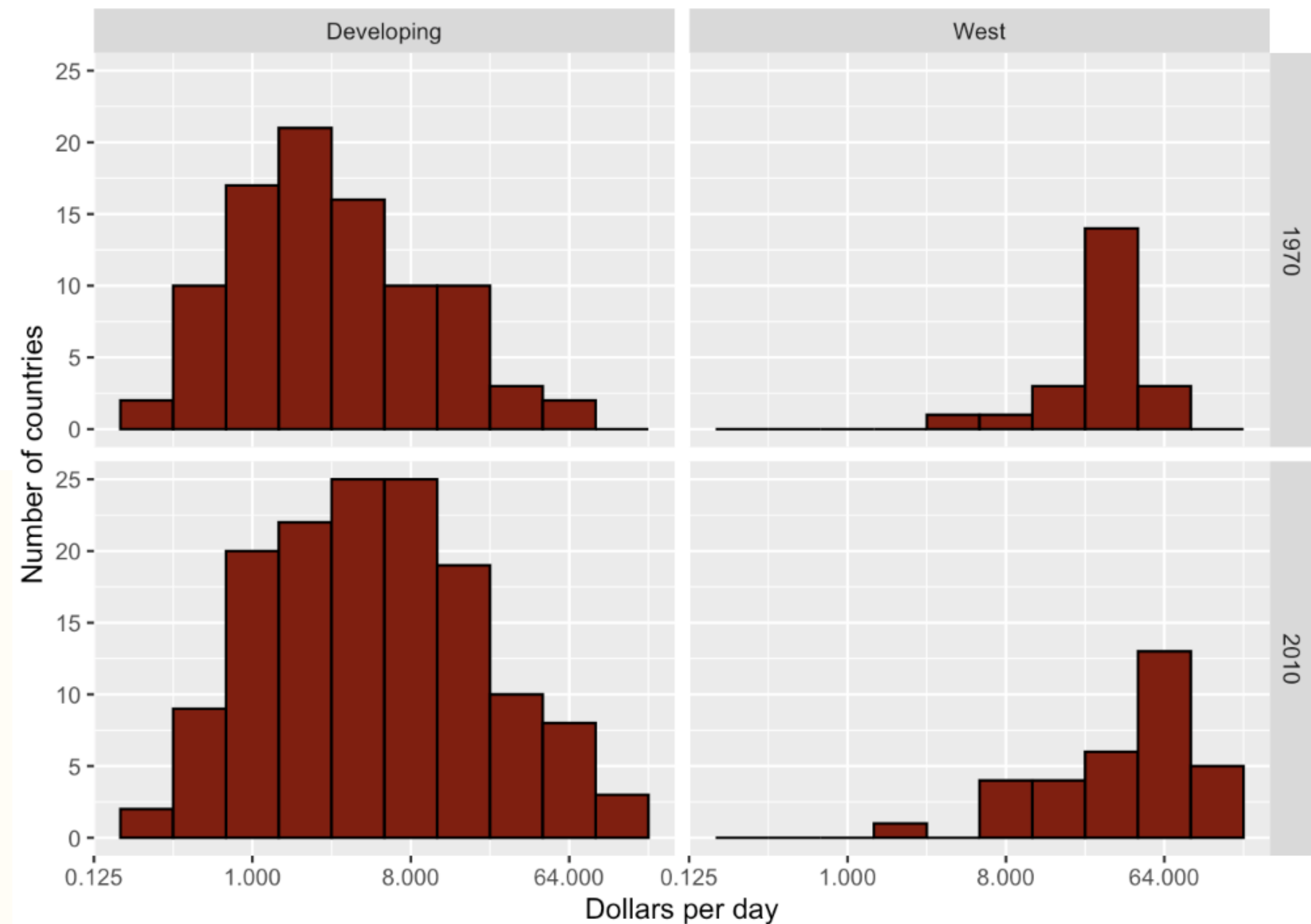
```
west <- c("Western Europe", "Northern Europe", "Southern Europe",  
         "Northern America", "Australia and New Zealand")
```

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_histogram(binwidth = 1, color = "black") +  
  scale_x_continuous(trans = "log2") +  
  facet_grid(. ~ group) +  
  xlab("Number of countries") +  
  ylab("Dollars per day")
```



Facet by two variables

```
past_year <- 1970
present_year <- 2010
gapminder %>%
  filter(year %in% c(past_year, present_year) & !is.na(gdp)) %>%
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%
  ggplot(aes(dollars_per_day)) +
  geom_histogram(binwidth = 1, color = "black") +
  scale_x_continuous(trans = "log2") +
  facet_grid(year ~ group) +
  xlab("Number of countries") +
  ylab("Dollars per day")
```



Make valid comparisons

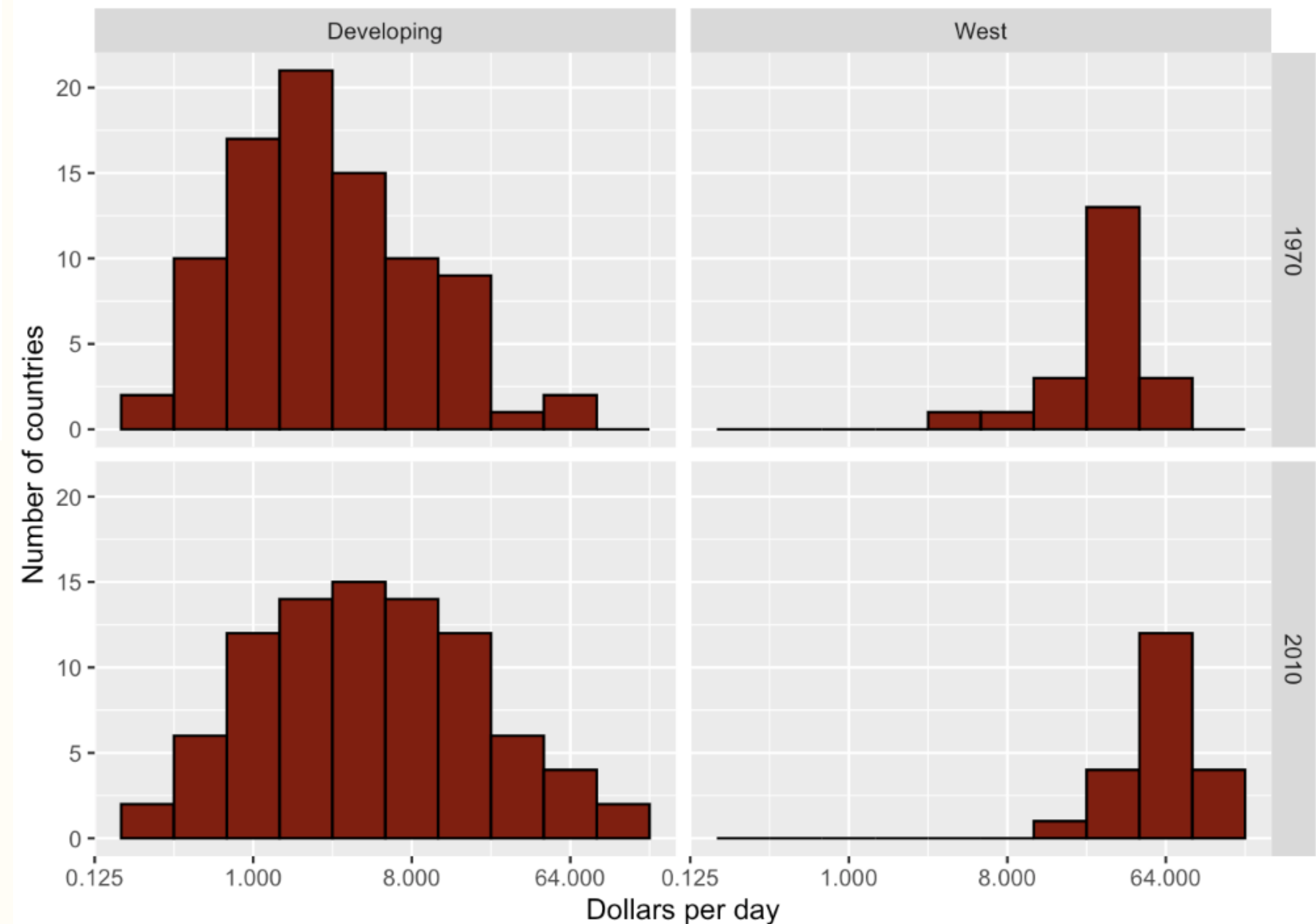
The number of countries represented in 1970 is less than the number in 2010

- More data for 2010
- Some countries split into multiple after 1970
- To make a **valid comparison** across years, we must have the **same countries for both years**

```
country_list_1 <- gapminder %>%  
  filter(year == past_year & !is.na(dollars_per_day)) %>%  
  .$country  
  
country_list_2 <- gapminder %>%  
  filter(year == present_year & !is.na(dollars_per_day)) %>%  
  .$country  
  
country_list <- intersect(country_list_1, country_list_2)
```

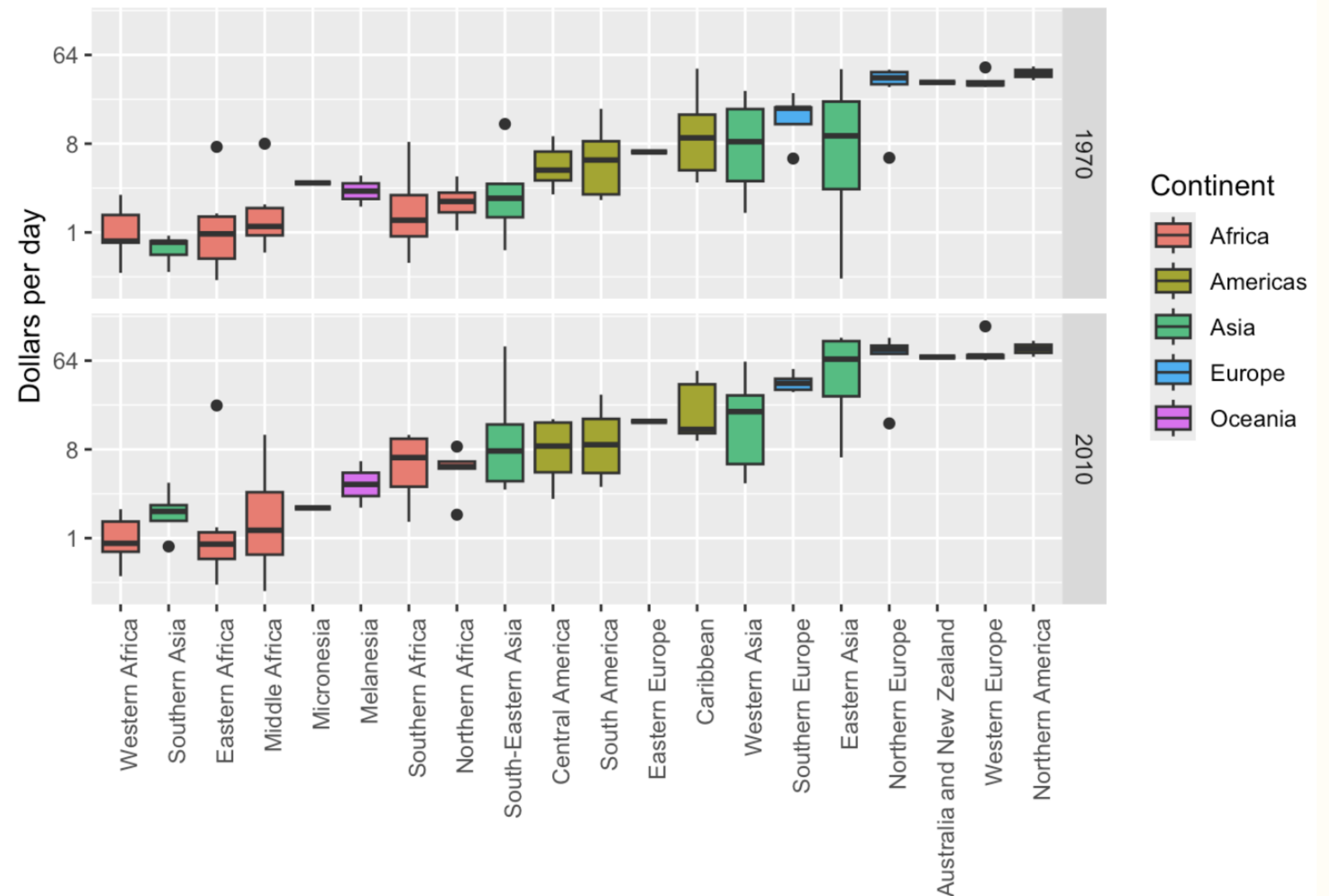
Make valid comparisons

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%
  ggplot(aes(dollars_per_day)) +
  geom_histogram(binwidth = 1, color = "black") +
  scale_x_continuous(trans = "log2") +
  facet_grid(year ~ group) +
  xlab("Number of countries") +
  ylab("Dollars per day")
```



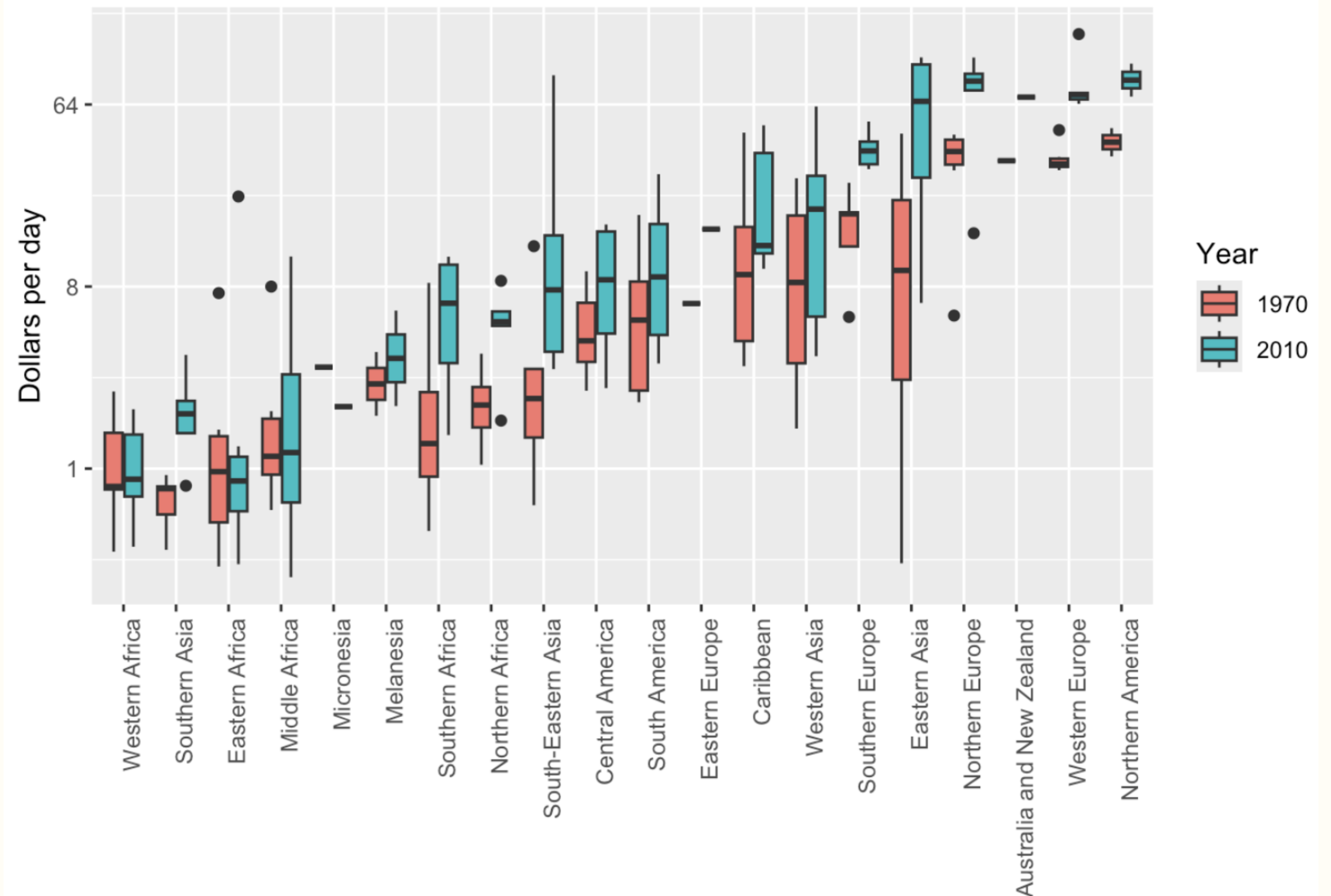
Boxplots across years

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%
  ggplot(aes(region, dollars_per_day, fill = continent)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  xlab("") +
  ylab("Dollars per day") +
  scale_y_continuous(trans = "log2") +
  labs(fill = "Continent") +
  facet_grid(year~.)
```



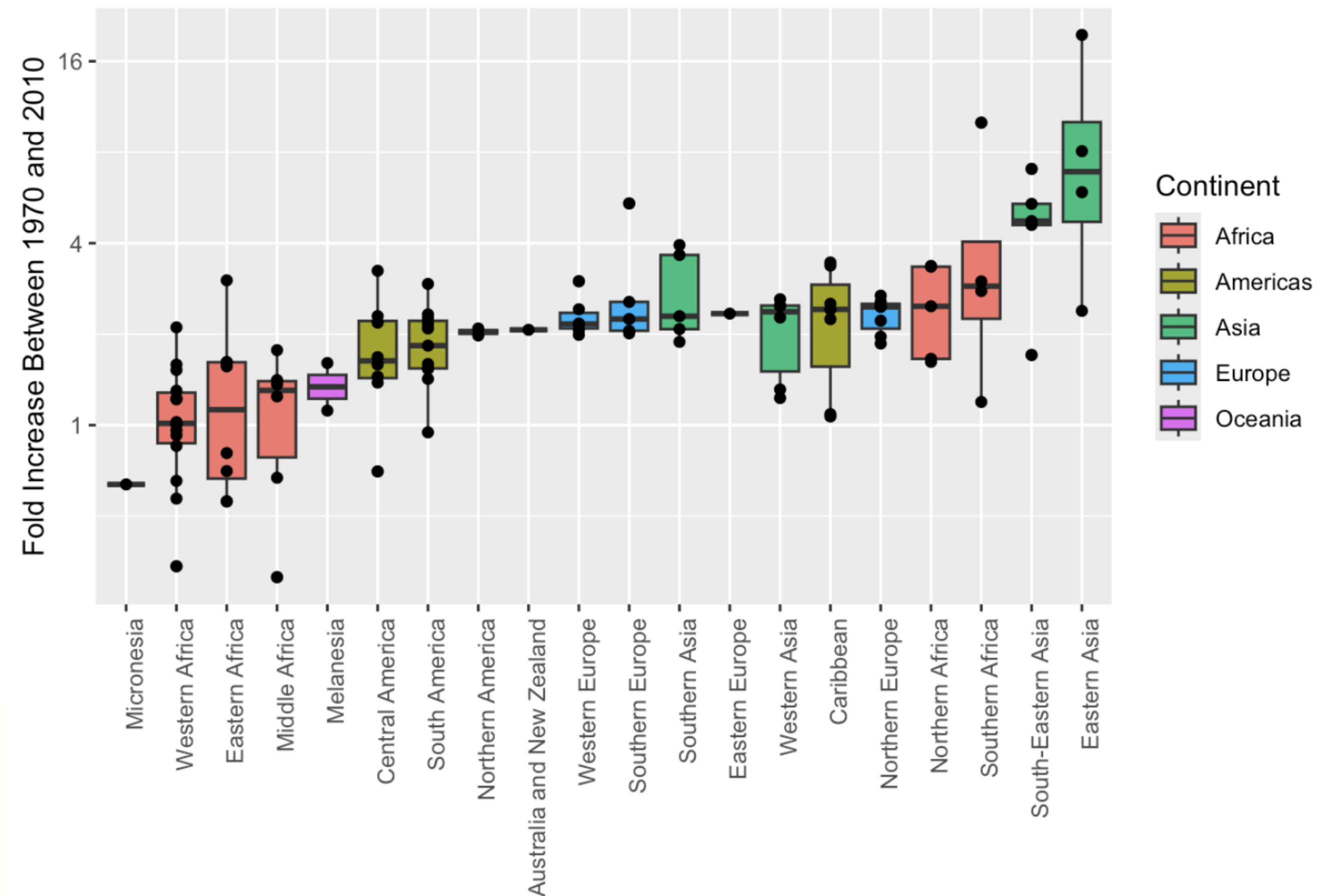
Plot side-by-side

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%  
  ggplot(aes(region, dollars_per_day, fill = factor(year))) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  xlab("") +  
  ylab("Dollars per day") +  
  scale_y_continuous(trans = "log2") +  
  labs(fill = "Year")
```



Fold change

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(year = ifelse(year == past_year, "past", "present")) %>%
  select(country, region, continent, year, dollars_per_day) %>%
  spread(year, dollars_per_day) %>%
  mutate(difference = present/past) %>%
  mutate(region = reorder(region, difference, FUN = median)) %>%
  ggplot(aes(region, difference, fill = factor(continent))) +
  geom_boxplot() +
  geom_point(show.legend = FALSE) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  xlab("") +
  ylab("Fold Increase Between 1970 and 2010") +
  labs(fill = "Continent") +
  scale_y_continuous(trans = "log2")
```



Summary

- Reordering by a meaningful value eases interpretation
- Comparisons of groups over time are only valid if the groups remain the same over time
- When comparing boxplots, plotting side-by-side rather than using faceting eases interpretation

Comparing distributions

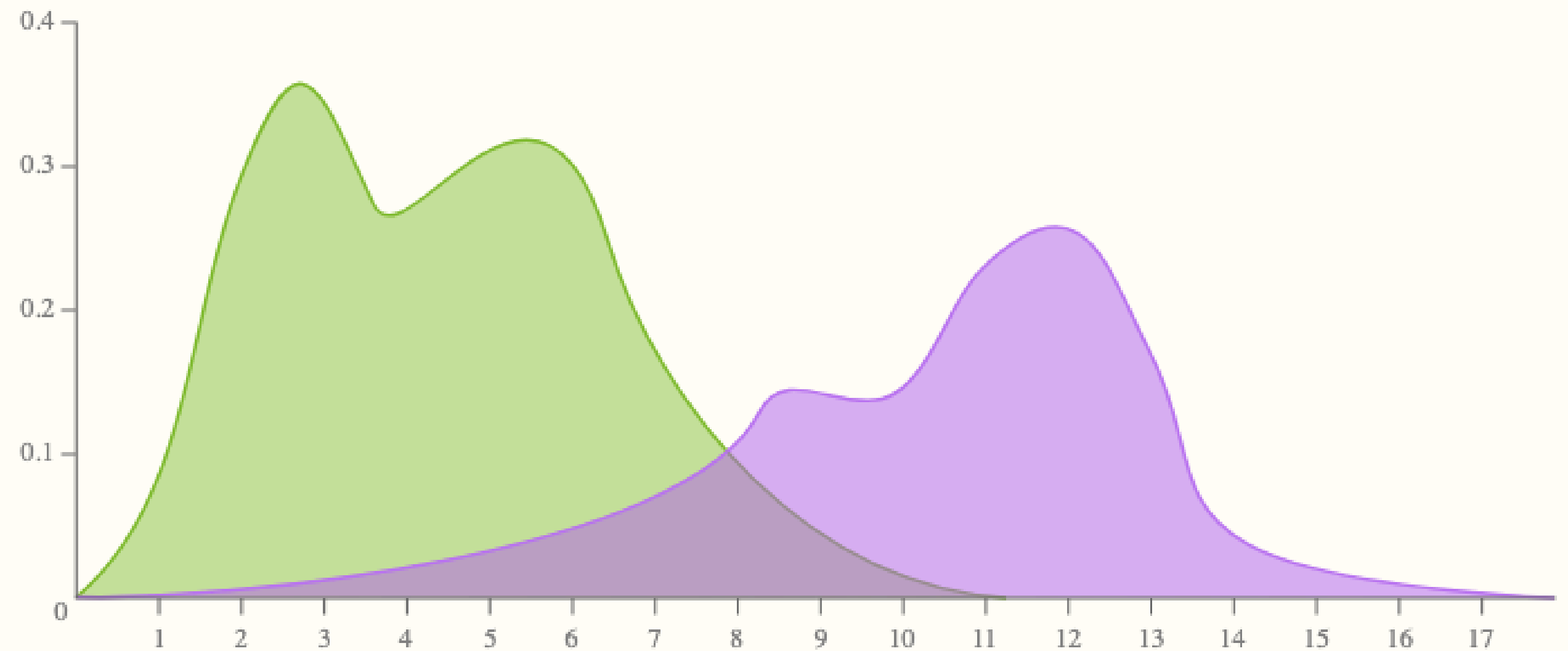
Smooth density plots

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Smooth density plots

Smooth density plots estimate the probability distribution of a continuous variable

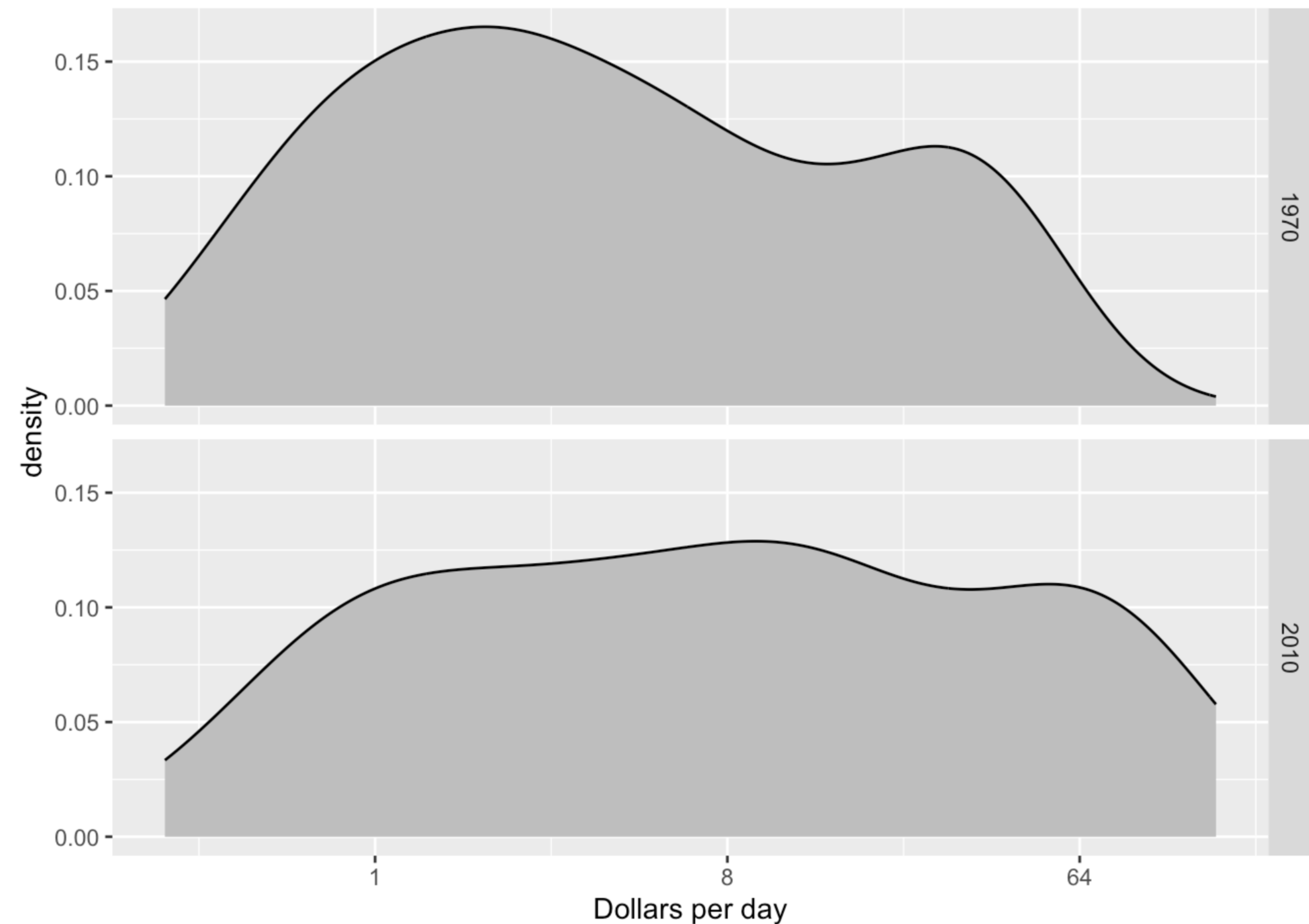
- A smoothed version of a histogram
- Useful for visualizing distribution shape
- Useful for comparing groups



Income distributions

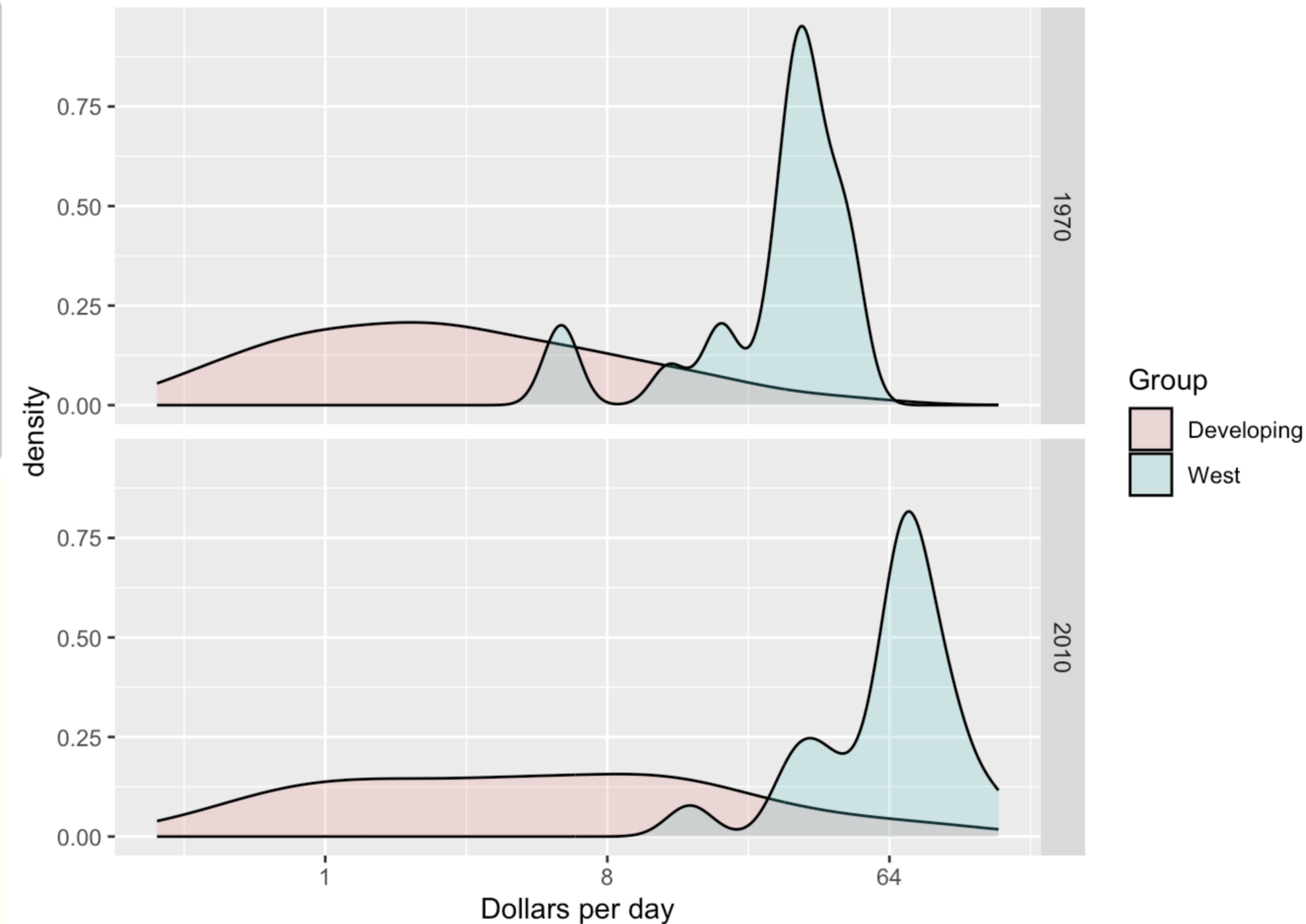
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
          country %in% country_list) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_density(fill = "grey") +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  facet_grid(year~.)
```

With this plot we can see that some countries have shifted to the right, closing the income gap – but we don't know **which countries** these are



Groups

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2) +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

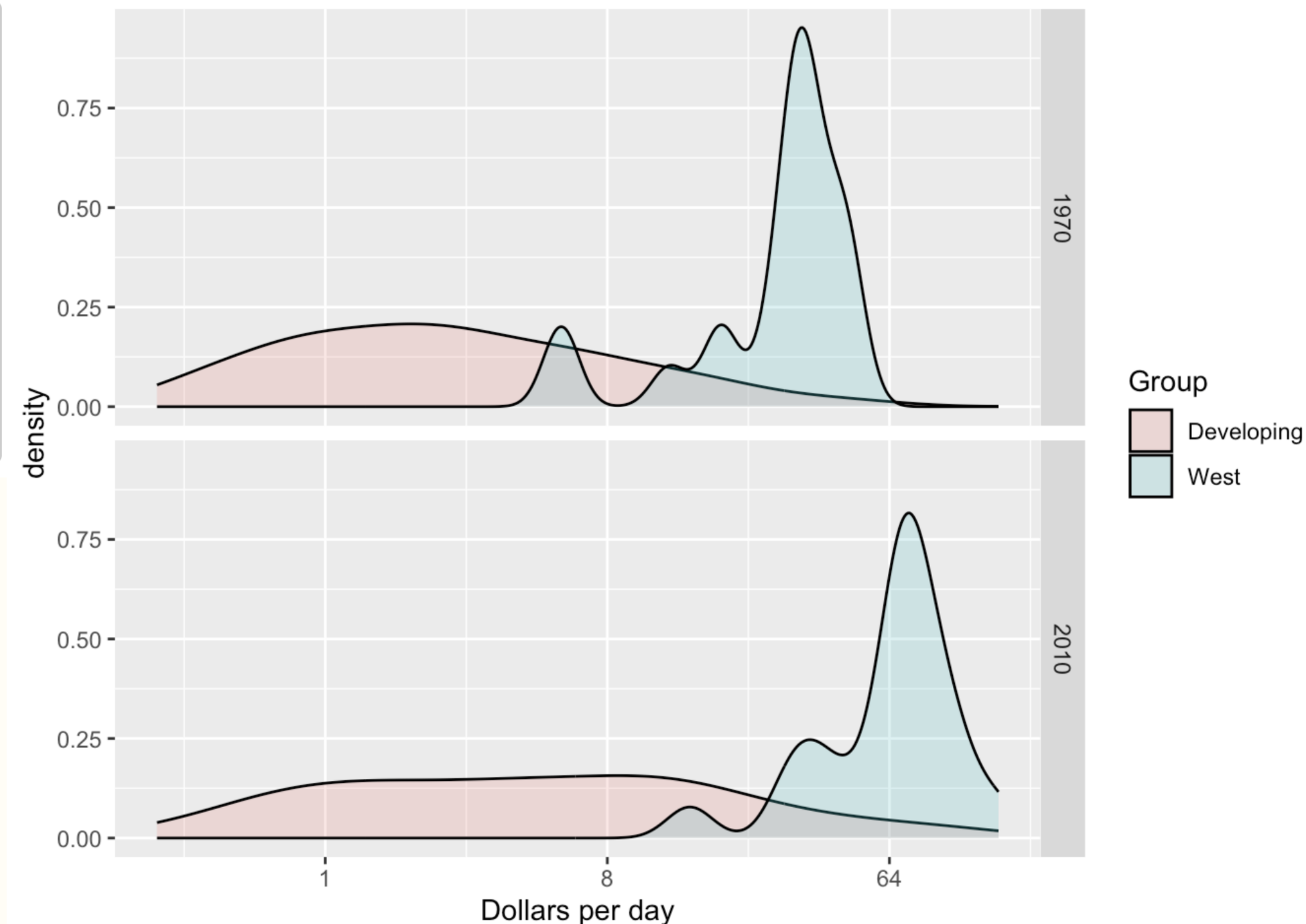


Groups

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%
  ggplot(aes(dollars_per_day, fill = group)) +
  scale_x_continuous(trans = "log2") +
  geom_density(alpha = 0.2) +
  facet_grid(year ~ .) +
  xlab("Dollars per day") +
  labs(fill = "Group")
```

The default is for the area under a smooth density plot to be **equal to 1**

This is not ideal in this context since there are 21 countries in our West group and 87 in our Developing group

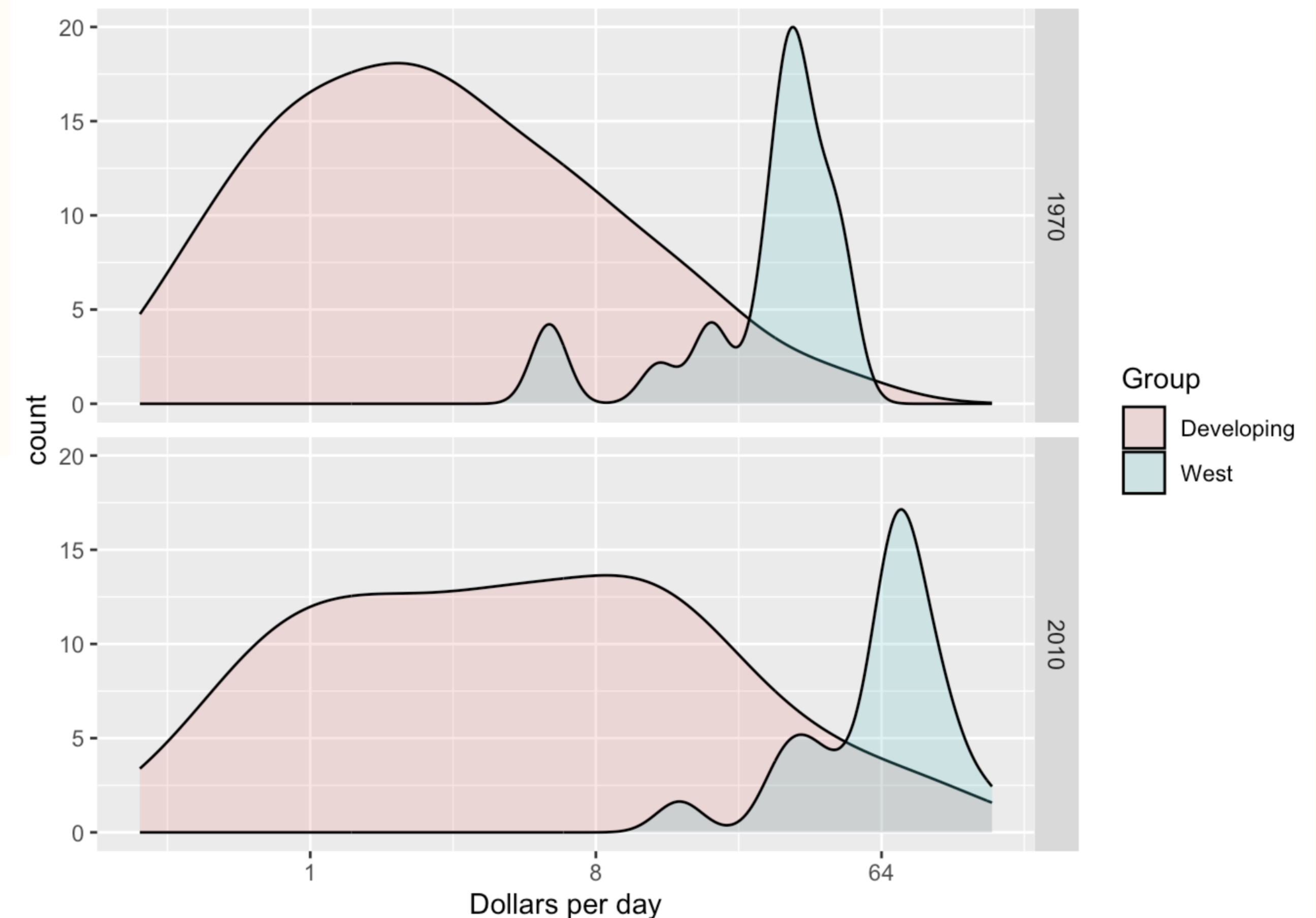


Accessing computed variables

Multiplying the y-axis values by the size of a group makes the area proportional to the number of countries in that group

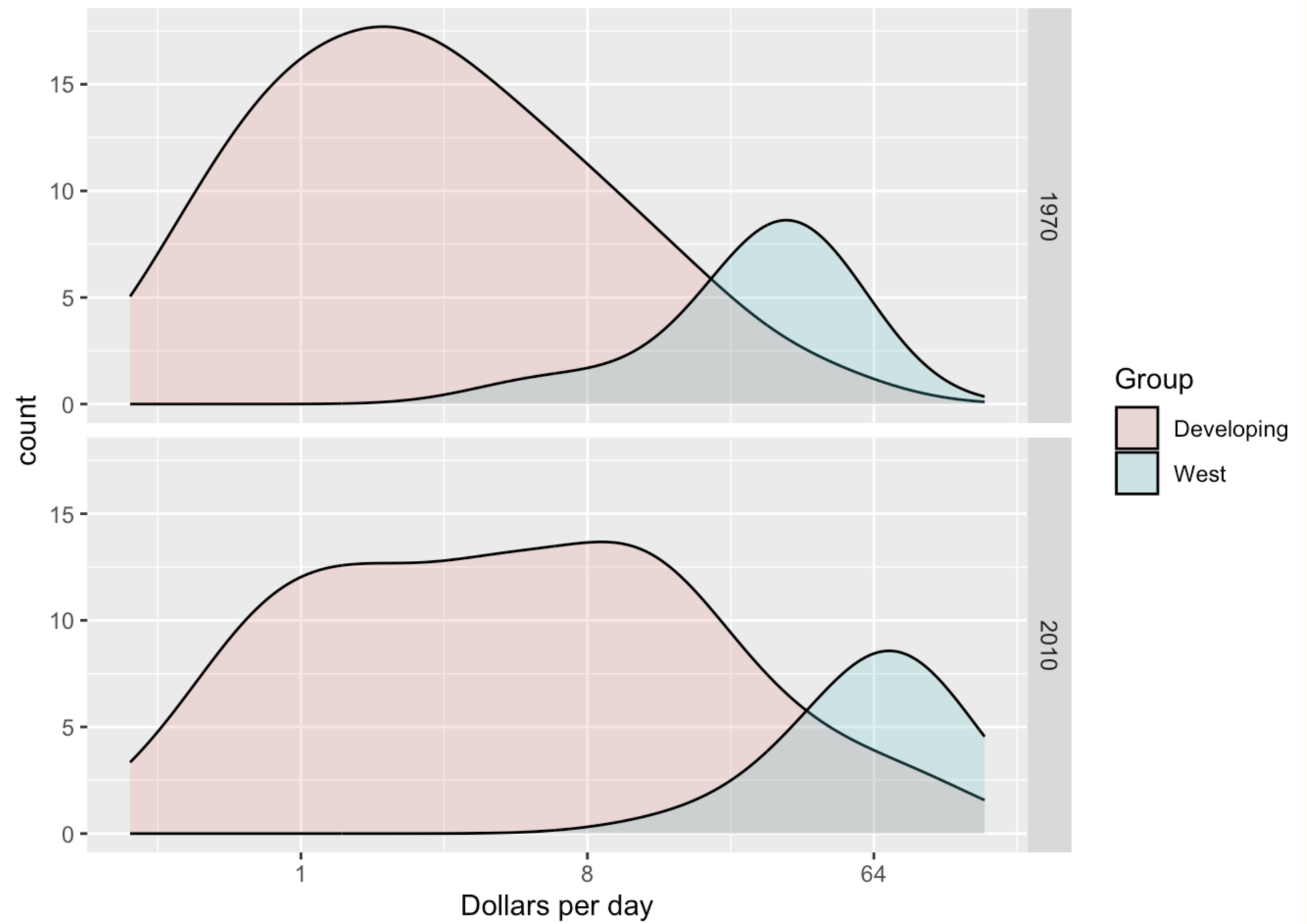
- **y = after_stat(count)**

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count), fill = group)) +  
  geom_density(alpha = 0.2) +  
  facet_grid(year ~ .) +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



Smoother densities (**bw**)

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count), fill = group)) +  
  geom_density(alpha = 0.2, bw = 0.75) +  
  facet_grid(year ~ .) +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



case_when

We can make this plot more informative by having more group categories

- The **case_when** function can create a new group variable efficiently

```
gapminder <- gapminder %>%  
  mutate(group = case_when(  
    region %in% west ~ "West",  
    region %in% c("Eastern Asia", "South-Eastern Asia") ~ "East Asia",  
    region %in% c("Caribbean", "Central America", "South America") ~ "Latin America",  
    continent == "Africa" & region != "Northern Africa" ~ "Sub-Saharan Africa",  
    TRUE ~ "Others"))
```

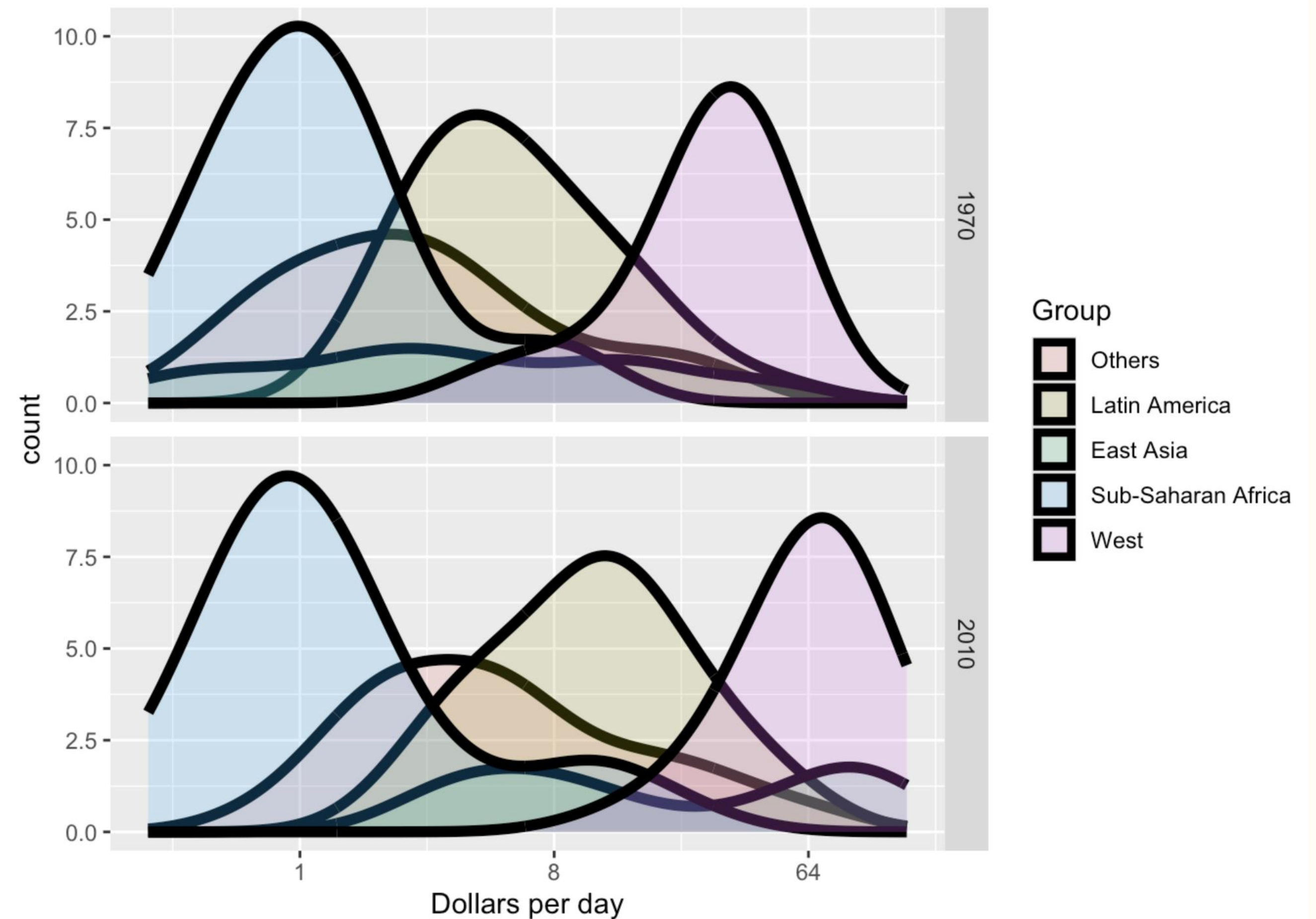
```
gapminder %>% select(region, group) %>% head()
```

##	region	group
## 1	Southern Europe	West
## 2	Northern Africa	Others
## 3	Middle Africa	Sub-Saharan Africa
## 4	Caribbean	Latin America
## 5	South America	Latin America
## 6	Western Asia	Others

linewidth

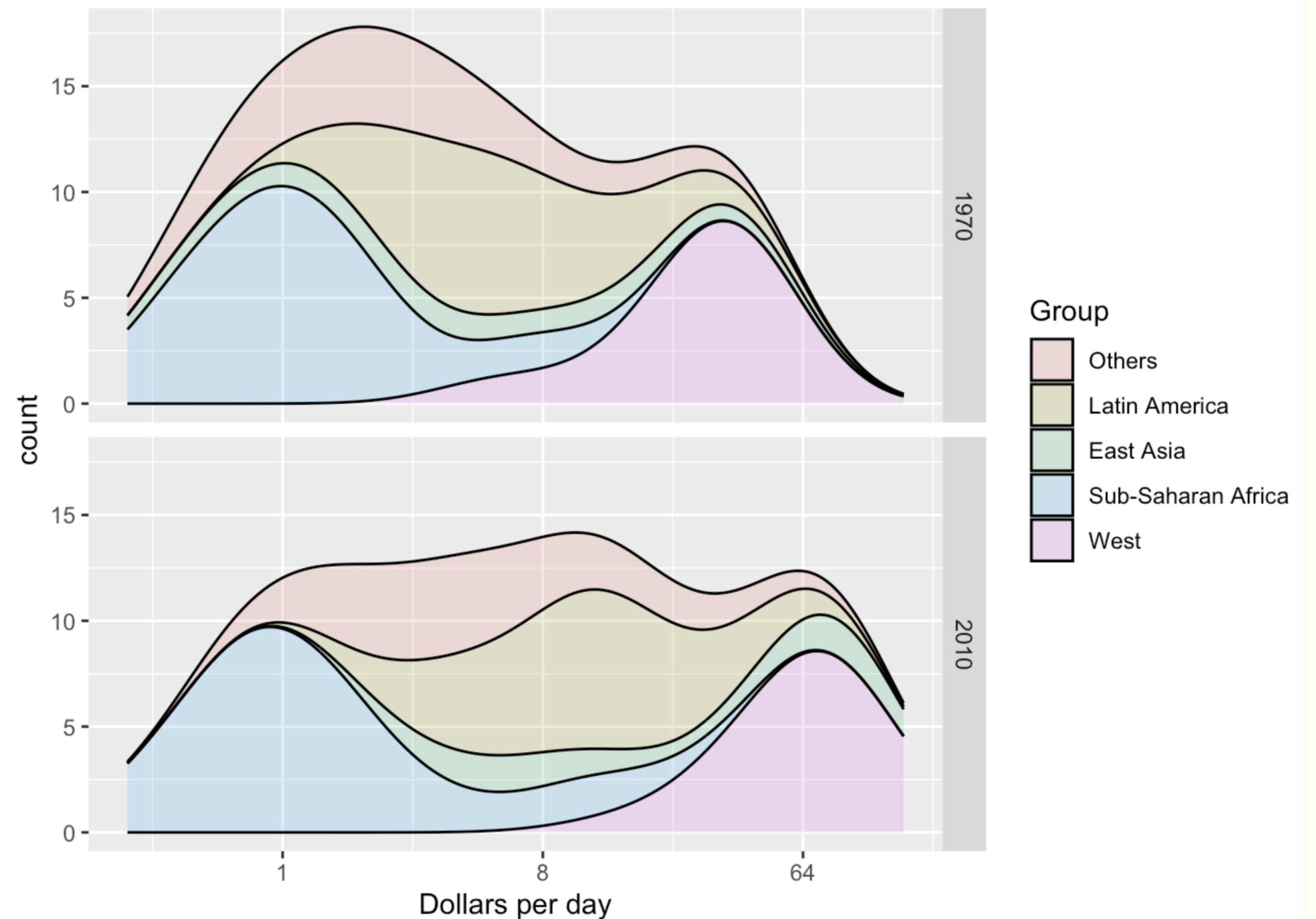
```
gapminder <- gapminder %>%  
  mutate(group = factor(group,  
    levels = c("Others", "Latin America",  
               "East Asia", "Sub-Saharan Africa",  
               "West")))
```

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
    scale_x_continuous(trans = "log2") +  
    geom_density(alpha = 0.2, bw = 0.75, linewidth = 2) +  
    facet_grid(year ~ .) +  
    xlab("Dollars per day") +  
    labs(fill = "Group")
```



position = "stack"

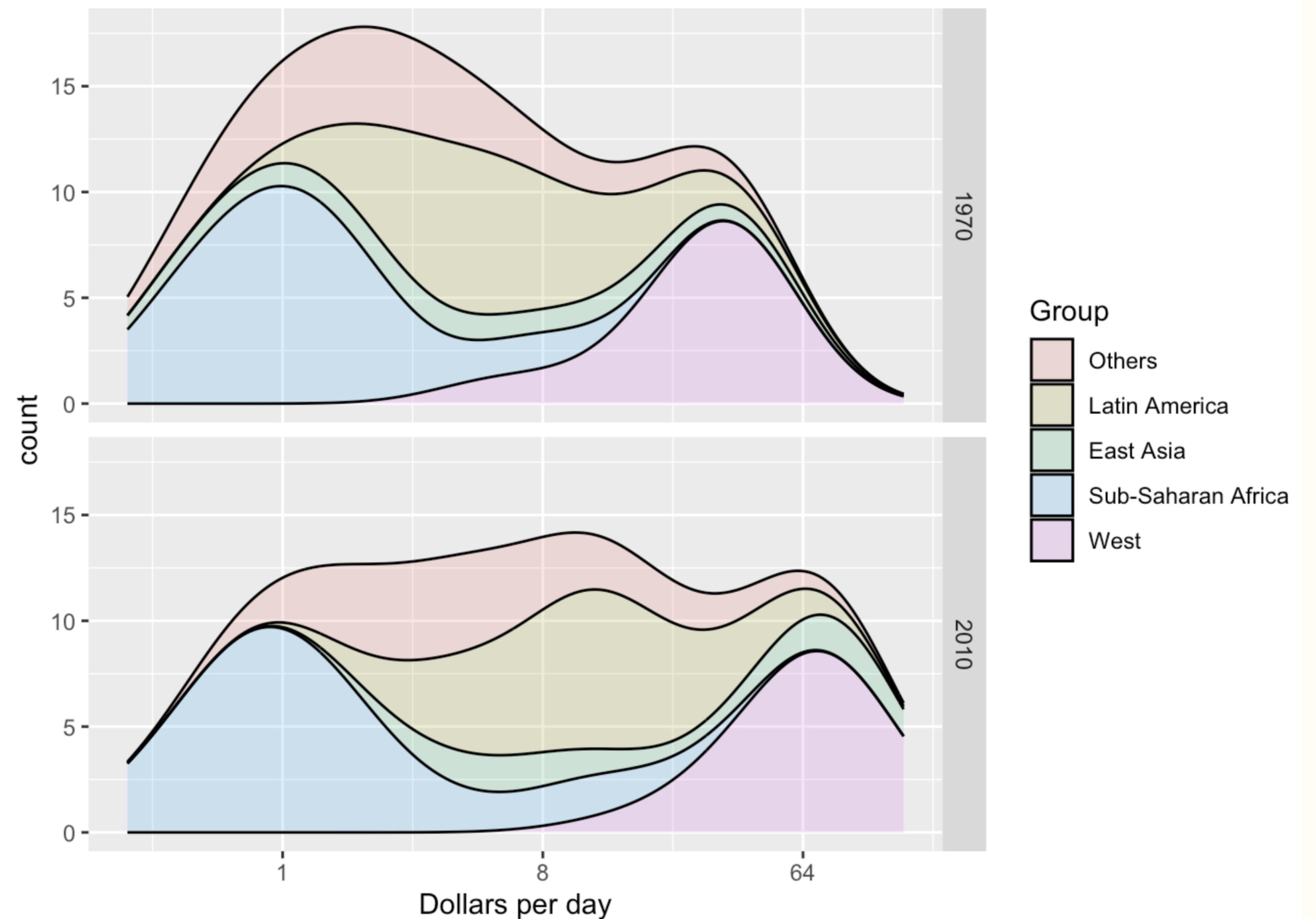
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



position = "stack"

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

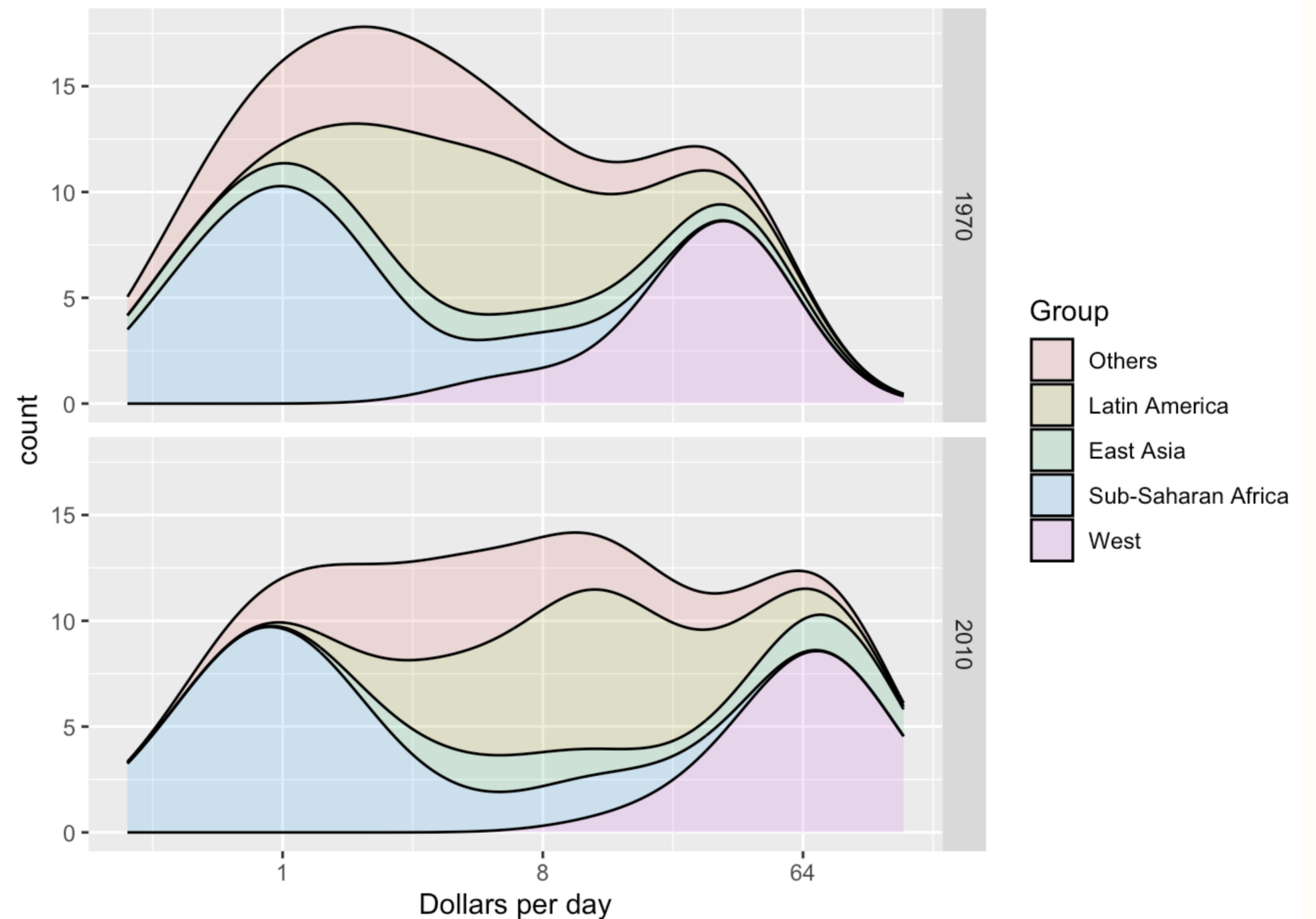
Plotting the two extremes (Sub-Saharan Africa and West) first shows the bimodality better



position = "stack"

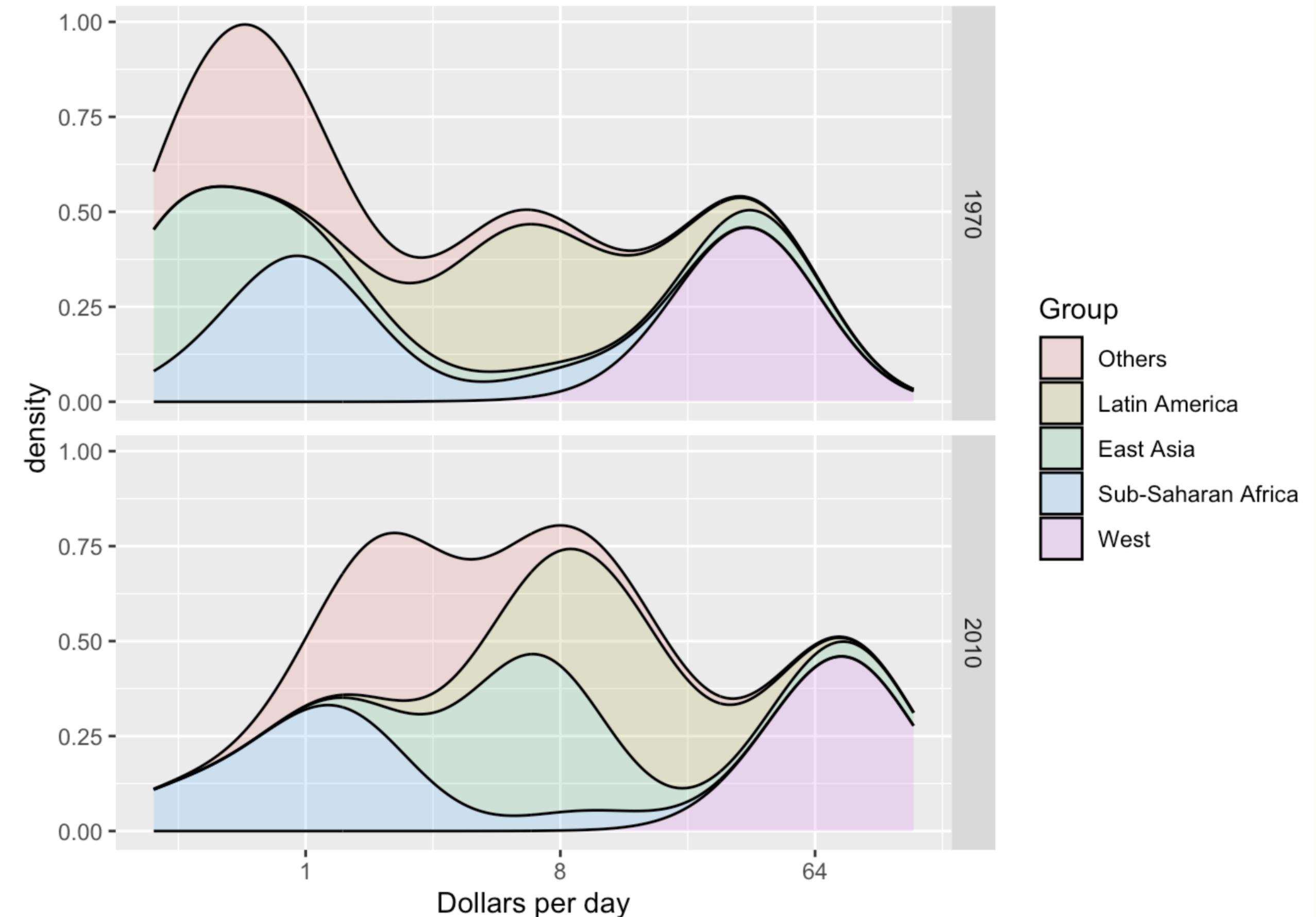
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

Here, each country has the same weight, which can skew comparisons – it's better to weight countries according to population



Weighted densities (**weight**)

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  group_by(year) %>%  
  mutate(weight = population/sum(population)) %>%  
  ungroup() %>%  
  ggplot(aes(dollars_per_day, fill = group, weight = weight)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



Summary

- Distributions across groups can be compared using smooth density plots
- The **case_when** function is an efficient way to create a categorical (factor) variable
- Smooth density plots can be stacked using the **position = "stack"** argument
- Smooth density plots can be weighted using the **weight** argument